

MEDDLE S.R.L.

CAU-7 Compressed-Air Unit

Instruction manual

Compressed-air production unit, oil-injected screw compressor with twin-tower desiccant dryer

Machine	CAU-7 Compressed-Air Unit (CAU-7)
Controller	CTRL-7
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1 Safety precautions

Read this chapter before you install, start or service the unit. Its rules apply everywhere in this book.

1.1 General rules

The CAU-7 Compressed-Air Unit produces, stores and dries compressed air for industrial installations. Use it for nothing else, and never feed its air to breathing equipment.

Only trained people may operate the unit, and only people who know its pressure and electrical hazards may service it.

Keep the enclosure panels closed and the guards in place while the motor turns. Do not hang pipework from the unit, alter its pressure envelope or change its wiring, and keep the cooling-air openings free; 5.1¹ states the room it needs. Obey national rules wherever they are stricter.

1.2 Pressure

The unit, the reservoirs and the pipework stay under pressure after the motor stops. Before you loosen a fitting or open the separator vessel:

- stop the unit at the controller and isolate the electrical supply;
- close the reservoir isolation valve;
- vent the unit and the reservoirs through the drain connection;
- confirm on the display that discharge pressure (P1) and reservoir pressure (P5) read zero.

Never exceed the maximum working pressure of 11.0 bar (160 psi). The safety valve on the separator vessel opens at 12.0 bar (174 psi) and is the last protection against overpressure: never plug, paint or readjust it. The reservoirs are built for 12.0 bar (174 psi); replace one that is dented, corroded or overheated.

Air at the purge silencer escapes as part of normal dryer operation; keep your face away, because the blast carries desiccant dust.

1.3 Electrical supply

The unit is fed from a three-phase supply of 400 V at 50 Hz; see 5.3². Only a qualified electrician may open the electrical box.

Isolate the unit at the external disconnecting device and lock it out before you open the box, work on the motor or remove a marked cover. The STOP key does not isolate.

The motor starts without warning. A unit standing still with its supply on is waiting to start: it starts as soon as line pressure (P2) falls to the cut-in pressure and the restart delay has run out. After a shutdown the start stays inhibited until the message has been reset.

1.4 Hot surfaces

The compression element, the separator vessel, the oil pipework, both coolers and the delivery pipe stay hot long after the unit has stopped. The controller warns before the oil reaches 85 °C (185 °F); see message X201 (3.4)³.

Let the unit cool before you drain the oil, change the oil filter or clean a cooler, and wear gloves and eye protection. Open the oil filler plug only when the unit is vented and cool: hot oil under pressure sprays.

1.5 Warning symbols

Three notices appear in this book and on the unit.

- **Danger** marks a hazard that causes death or serious injury.
- **Warning** marks a hazard that can injure.
- **Caution** marks a practice that can damage the unit or spoil the delivered air.

A symbol on the unit repeats the notice where the hazard is: electrical at the electrical box, hot surface at the compression element, pressure at the separator vessel and the reservoirs. Replace an unreadable label. The controller raises messages of its own; 3.3⁴ explains each type.

1. Section 5.1, Site
2. Section 5.3, Electrical connection
3. Section 3.4, Message list
4. Section 3.3, Message types

2 General description

This chapter describes what the unit is made of and how the parts work together. Read it once before you operate the unit; the procedures in the later chapters assume that you know the names used here.

2.1 What the unit does

The CAU-7 Compressed-Air Unit is a complete compressed-air station on one frame. An oil-injected single-stage screw compressor driven by a fixed-speed motor produces the air. An aftercooler and a cyclonic separator remove the bulk of the condensate, a twin-tower heatless desiccant dryer removes the rest, and two air reservoirs store the dried air and damp the demand of the installation. A CTRL-7 controller regulates the unit, watches its sensors and reports what it finds. A sound-insulated enclosure surrounds the whole assembly.

At the reference conditions of 5.2⁵ the unit delivers 0.45 m³/min (16 cfm) at its maximum working pressure of 11.0 bar (160 psi). The motor takes 4.0 kW (5.4 hp) and turns at 2950 rpm. The complete ratings are listed in 9.1⁶; this chapter explains how the unit reaches them, and 2.7⁷ shows every part in one picture.

2.1.1 Main assemblies

On the compressor side the air intake filter, the intake and unloading valve, the screw compression element, the drive motor and the motor coupling sit together on the frame. The oil separator vessel carries the separator element, the minimum-pressure valve, the safety valve and the oil level switch, and the blow-down valve vents it.

The cooling group holds the oil cooler, the aftercooler, the cooling fan and the thermostatic valve; the oil filter sits between the cooler and the compression element. Downstream, the cyclonic separator and its automatic condensate drain feed desiccant tower 1 and desiccant tower 2, which the dryer changeover valves, the dryer purge valve and the purge silencer serve. The air reservoirs, the reservoir inlet flow sensor and the reservoir isolation valve close the air path, and the pneumatic panel carries the line-pressure instruments. The CTRL-7 controller sits behind the front door.

2.2 Air flow

Air enters through the air intake filter, which keeps dust out of the compression element, and passes the intake and unloading valve. That valve opens only while the controller has loaded the unit; at every other moment it is closed, and the unit draws no air.

The screw compression element compresses the air and injects oil into it, so what leaves the element is a mixture of air and oil mist. The mixture enters the oil separator vessel, where most of the oil drops out by gravity and change of direction. The separator element in the top of the vessel catches the mist that remains. discharge pressure (P1) is measured in the vessel, and oil temperature (T1) at the outlet of the compression element.

The minimum-pressure valve at the vessel outlet is a non-return valve with a spring. It holds enough pressure in the vessel to keep the oil circulating, and it stops air in the delivery line from flowing back when the unit unloads. The valve opens at 4.0 bar (58 psi), so the vessel is never emptied faster than the separator element can drain.

Past the valve the air runs through the aftercooler, where it gives up most of its heat, and into the cyclonic separator, which spins out the condensate that the cooling has produced. The automatic condensate drain empties the separator into the condensate connection, and separator discharge pressure (P3) is measured at the separator discharge port.

The cooled and drained air then passes the dryer described in 2.5⁸, flows through the reservoir inlet flow sensor into the two air reservoirs, and leaves through the reservoir isolation valve to the pneumatic panel. There line pressure (P2) and the low-pressure switch watch the pressure the installation actually sees, and reservoir pressure (P5) is measured immediately downstream of the reservoirs.

2.3 Oil circuit

The oil does three jobs inside the compression element: it seals the clearance between the rotors, it carries away the heat of compression, and it lubricates the bearings. The unit holds 8.0 L (2.1 US gal) of synthetic screw compressor oil.

There is no oil pump. Air pressure in the oil separator vessel drives the circuit: oil collects in the bottom of the vessel and is pushed out to the thermostatic valve, which sends it either straight back to the compression element or through the oil cooler, depending on how warm it is. From there the oil passes the oil filter, which holds back wear particles, and re-enters the compression element.

A scavenge line returns the oil that collects under the separator element to the low-pressure side of the compression element. The line carries very little oil and is easily blocked; a blocked scavenge line is the usual reason for oil in the delivered air.

The oil level switch in the separator vessel reports whether the level is above the minimum mark. Read the level from the sight glass as part of [task 7.3.1⁹](#), with the unit stopped and vented.

2.4 Cooling

One cooling fan draws ambient air through the enclosure and pushes it across the oil cooler and the aftercooler. The same stream cools the drive motor and the electrical box, so the unit needs a supply of cool air even when the oil itself is not yet warm. ambient temperature (T2) is measured at the cooling-air inlet.

The thermostatic valve keeps the oil out of the cooler until the oil is warm enough. That matters in a cold room: oil that is too cold is too thick to circulate, and condensate then collects in the separator vessel instead of evaporating. The unit is built to run with cooling air between 2 °C (36 °F) and

5. Section 5.2, Reference conditions

6. Section 9.1, Technical data

7. Section 2.7, System schematic

8. Section 2.5, Air drying

9. Section 7.3.1, Daily checks

40 °C (104 °F). Outside that window the controller reports an ambient message, and the oil temperature soon follows it. 5.1¹⁰ states what the window means for the room the unit stands in.

2.5 Air drying

The dryer is a twin-tower heatless desiccant dryer. Each tower holds a charge of desiccant. One tower dries the air that flows to the reservoirs while the other one regenerates, and the changeover valves swap their roles at the end of every tower period. dryer tower in service (D3) tells you which of the two is drying.

Regeneration needs no heater. A part of the already dried air is expanded to atmospheric pressure and led backwards through the off-line tower, where it picks up the water the desiccant collected during its drying period and carries it out through the dryer purge valve and the purge silencer. That purge air is 15 % of the delivered flow, which is why the unit compresses more air than the installation consumes.

dryer purge pressure (P4) is measured in the purge line downstream of the purge valve. It is near zero while a tower regenerates normally and rises only in short pulses at changeover. A purge pressure that stays high means that air is escaping continuously, either through a purge valve that no longer seats or through a damaged silencer. purge pressure switch (D6) confirms that the off-line tower really discharges.

The tower period is 60 s, and the changeover is linked to demand rather than to the clock: the towers change over only while the unit is delivering air, because only then is there dry air to regenerate with. After every cut-in, tower 1 takes the first period, so a unit that cycles frequently keeps returning to the same tower. At the reference conditions the dryer reaches a pressure dew point of -40 °C (-40 °F). 6.5¹¹ describes what the operator sees during a changeover.

2.6 Regulation

The unit runs at fixed speed and is regulated by loading and unloading. While the unit is loaded, the intake valve is open and the compression element delivers air. When line pressure (P2) reaches the cut-out pressure of 10.0 bar (145 psi), the controller de-energises the load solenoid: the intake valve closes, the blow-down valve vents the separator vessel, and the motor keeps turning without delivering air. When the pressure falls back to the cut-in pressure of 8.0 bar (116 psi), the controller loads the unit again.

A unit that has unloaded keeps running for the unloaded run-on time, so that a short pause in demand does not cost a motor start; see parameter P03 (4.2)¹² for the parameter and chapter 6¹³ for its value and its effect. If no load request arrives within that time, the motor stops and the unit waits in the off state. A restart delay then keeps the motor off for a moment even if the pressure already calls for air, which protects the motor against short start intervals.

The pneumatic panel also carries a hardware low-pressure switch that works independently of the controller. It closes below 7.0 bar (102 psi) and opens again above 7.7 bar (112 psi), so the controller learns that the installation has lost pressure even if the line-pressure transducer is faulty. 6.3¹⁴ shows what a complete load cycle looks like.

2.6.1 Panel labels

Every measured value carries a short panel label, and the display, the schematic in this chapter and the signal list all use the same one: the pressures P1 to P5, the temperatures T1 and T2, the motor current I1 and the digital inputs D1 to D8. A label tells you where a value is measured, not what it should read; 9.2¹⁵ gives the complete list and 9.3¹⁶ the band each value normally stays in.

2.7 System schematic

10. Section 5.1, Site
11. Section 6.5, Dryer operation
12. Section 4.2, Parameter list
13. Chapter 6, Operation
14. Section 6.3, The normal load cycle
15. Section 9.2, Signal list
16. Section 9.3, Normal bands

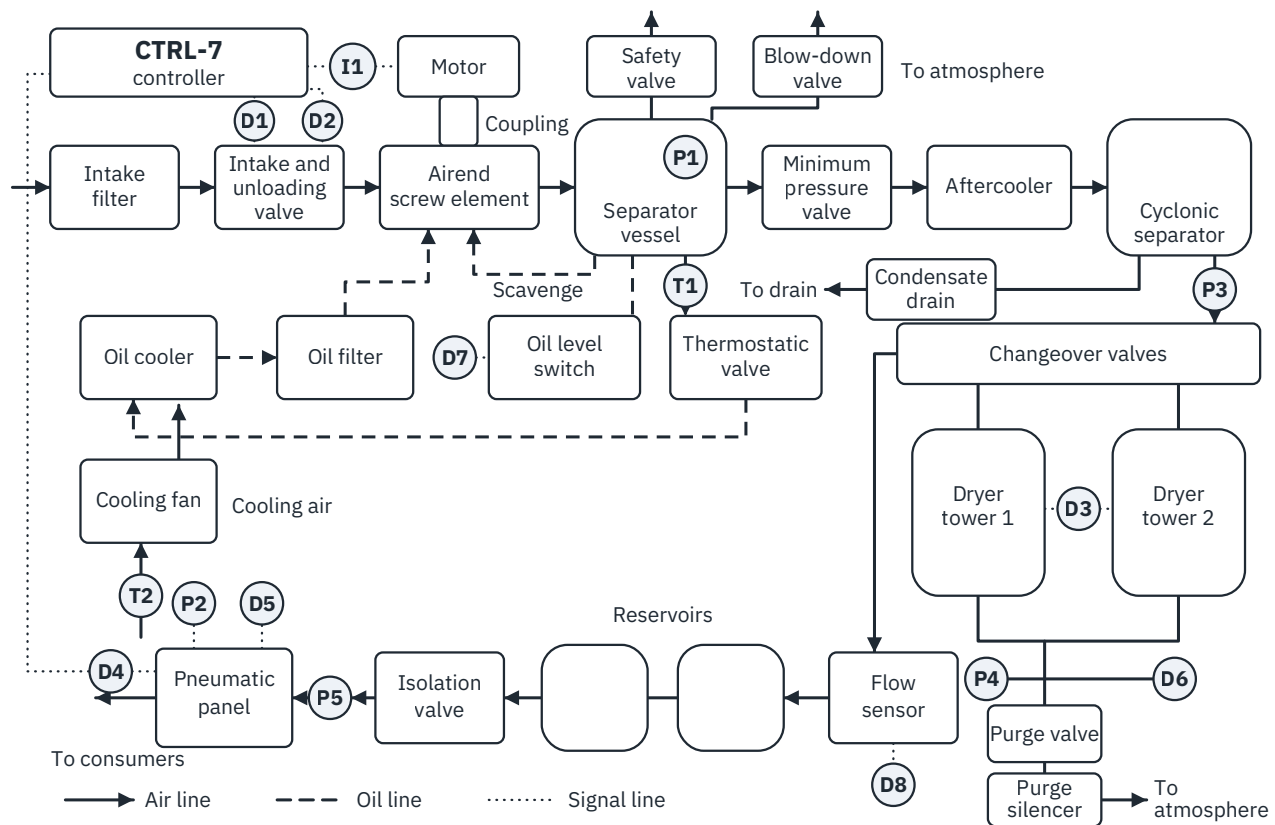


Figure 2.1 – Air, oil and signal flow of the CAU-7

The schematic follows the air from the intake filter on the left to the consumers on the right, with the oil circuit drawn as the loop that returns to the compression element and the controller shown as the box that collects the measured values. Every instrument is drawn at the place where it is installed and carries its panel label, so the picture and the readings on the display use the same names. The two desiccant towers are drawn side by side with the changeover valves between them and the purge path leading down to the silencer. Use the schematic together with 9.2¹⁷ when you trace a fault: it is the quickest way to see which part sits upstream of the reading that surprised you.

3 Controller CTRL-7

3.1 Display

The CAU-7 is operated from the CTRL-7 controller in the front panel. It regulates the unit, supervises every measured signal, counts the running hours behind the service intervals and holds the programmable settings. Everything an operator has to see, acknowledge or change is on the panel.

The display has two lines and shows one of two pictures.

- While no message is active, the upper line shows line pressure (P2), the value the unit regulates on, and the lower line shows the machine state — off, unloaded or loaded — with the run-hour counter.
- While a message is active, the upper line carries its code and the lower line the short text that belongs to that code. Both are listed in 3.4¹⁸.

When several messages are active, the most severe is shown first: shutdown, shutdown warning, warning, service message. Step through the rest with the arrow keys.

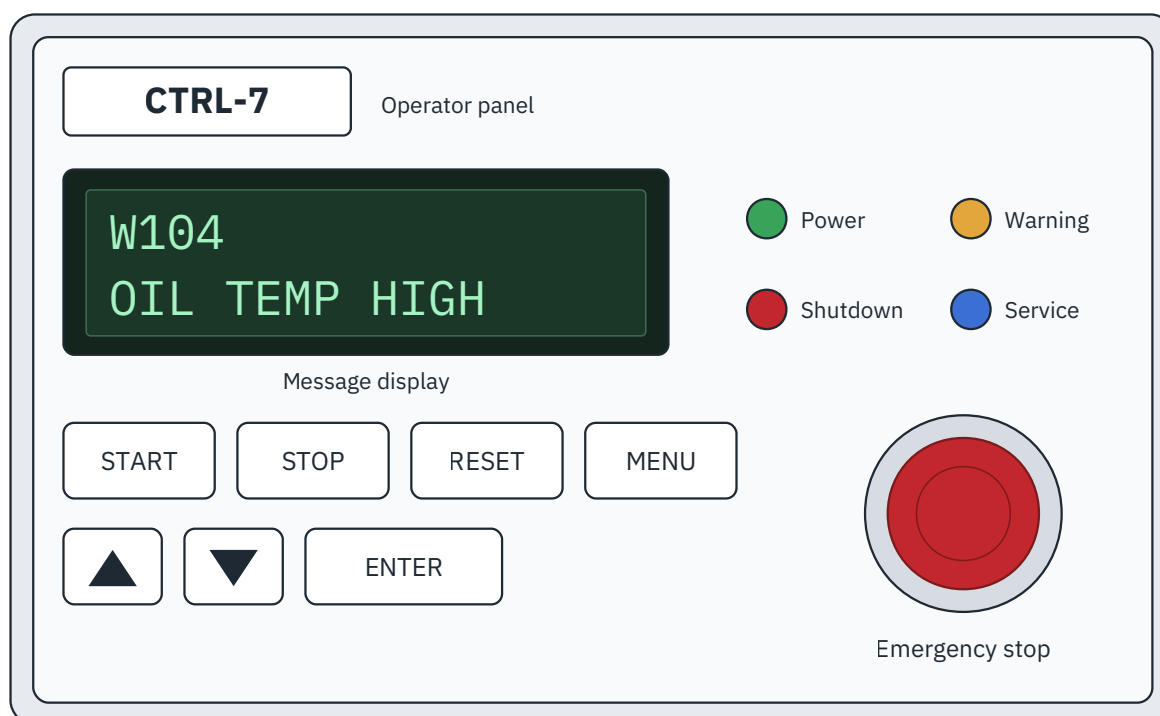


Figure 3.1 — CTRL-7 front panel

The figure shows the panel with the oil temperature warning active. Paging on reaches the measured signals, so the reading behind a message can be read without acknowledging it.

3.2 Keys and indicator lamps

- **START** starts the unit through the sequence of 6.1¹⁹. It is refused while a shutdown is active.
- **STOP** stops the unit through the unload and run-on sequence of 6.2²⁰. Always stop with this key, never by switching off the supply.
- **RESET** acknowledges the message on the display. A message whose condition remains reappears at once.
- **MENU** opens and leaves the settings menu of 4.1²¹.
- The **arrow keys** page through the messages and the measured signals, select a parameter and change a value.
- **ENTER** confirms a selection or a changed value.
- The **emergency-stop button** is wired into the motor circuit; it is not a controller key and works whatever the controller is doing.

18. Section 3.4, Message list

19. Section 6.1, Starting the unit

20. Section 6.2, Stopping the unit

21. Section 4.1, Reaching the settings

Four lamps repeat the state of the unit for anyone too far away to read the display. **Power**, green, is lit whenever the controller is supplied; **warning**, amber, while a warning is active, flashing while a shutdown warning is; **shutdown**, red, while the unit is stopped by a shutdown; **service**, blue, while a service message is active.

3.3 Message types

Every message carries a code whose first letter gives the type and decides what the unit does. The number behind the letter only groups the messages of that type.

- **W — warning.** A value has left its limit, or a signal is missing. The unit keeps running. Deal with the cause at the next opportunity; an ignored warning usually becomes a shutdown warning.
- **X — shutdown warning.** The same quantity has moved further and is close to a shutdown limit. The unit still runs, but the controller will stop it if the value keeps moving. Act at once: reduce the demand or remove the cause.
- **S — shutdown.** The controller has stopped the unit to protect it, and start stays inhibited until the message is reset. Never bridge, disable or repeatedly reset a shutdown to keep production running.
- **M — service.** A run-hour counter or a calendar interval has expired. The unit keeps running and only the service lamp lights.

3.4 Message list

The list holds every message the CTRL-7 can show: the code, the type, the display text, the plain title, the threshold and delay that make the message appear, and what the unit does while it is active. The causes behind a message, and the checks that separate them, are in the problem-solving chapter.

Controller messages

Code	Message	Trigger	Threshold	Delay	Reset	Action
W101	Line pressure low LINE PRESS LOW	low-pressure switch (D5) equal to the limit in state any	1	10 s	automatic	Switch off the consumers you can spare and watch whether line pressure (P2) recovers. If it does not, work through section 8.3 .
W102	Continuous load time exceeded CONT LOAD TIME	continuous load time above the limit while loaded	10 min	0 s	automatic	Check whether the demand is genuine. If no consumer explains it, look for escaping air at the purge silencer and in the distribution system, then follow section 8.5 .
W103	Dryer purge pressure high PURGE PRESS HIGH	dryer purge pressure (P4) above the limit while loaded	0.5 bar (7 psi)	60 s	automatic, with hysteresis (0.1 bar)	Listen at the purge silencer for a permanent discharge and, if you hear one, arrange task 7.3.10 .
W104	Oil temperature high OIL TEMP HIGH	oil temperature (T1) above the limit while the unit runs	75 °C (167 °F)	300 s	automatic, with hysteresis (5 °C)	Check that the cooling air path is free and that the fan runs, then read the oil level while the unit is stopped. The unit keeps running, but the shutdown limit is not far above this warning.
W105	Oil temperature low OIL TEMP LOW	oil temperature (T1) below the limit in state any	5 °C (41 °F)	60 s	automatic	Heat the machine room before starting the unit, and have the thermostatic valve and the temperature sensor checked if the reading stays low once the unit is warm.
W106	Motor current high MOTOR CURR HIGH	motor current (I1) above the limit while loaded, outside the start mask	7.5 A	60 s	automatic	Note the pressure at which the current rises and compare it with the values in section 9.3 , then follow section 8.9 .
W107	Motor current low under load MOTOR CURR LOW	motor current (I1) below the limit while loaded, outside the start mask	5.0 A	60 s	automatic	Inspect the intake filter and arrange task 7.3.3 if it is dirty, then follow section 8.10 .
W108	Motor starts per hour exceeded TOO MANY STARTS	motor starts per hour above the limit in state any	6 /h	0 s	automatic	Check that both reservoirs are in service and look for escaping air in the distribution system, then follow section 8.4 .
W109	Ambient temperature high AMBIENT T HIGH	ambient temperature (T2) above the limit in state any	40 °C (104 °F)	600 s	automatic	Clear the ventilation openings and check that the extraction works, keeping the site requirements of section 5.1 in mind.
W110	Ambient temperature low AMBIENT T LOW	ambient temperature (T2) below the limit in state any	2 °C (36 °F)	600 s	automatic	Heat the machine room to the range given in section 5.1 before starting the unit, and check that the condensate drain is not frozen.

Code	Message	Trigger	Threshold	Delay	Reset	Action
W111	Dryer changeover not detected NO TOWER CHANGE	seconds since tower change above the limit while loaded	180 s	0 s	automatic	Watch the dryer through a full cycle and listen for the regeneration discharge, then follow section 8.18.
W112	Separator discharge pressure high under load SEP PRESS HIGH	separator discharge pressure (P3) above the limit while loaded	0.5 bar (7 psi)	60 s	automatic	Watch the drain over a few cycles to see whether it opens and closes, and arrange task 7.3.8 if it does not.
W113	Reservoir pressure deviation RESERVOIR DEV	reservoir line delta above the limit in state any	0.5 bar (7 psi)	60 s	automatic	Check that the reservoir isolation valve is fully open, then compare both readings on the controller as described in section 8.17.
W114	Flow signal lost NO FLOW SIGNAL	flow pulse (D8) equal to the limit while loaded	0	120 s	automatic	Confirm that air actually reaches the consumers, then have the flow sensor and its wiring checked; see section 8.19.
W115	Purge switch not detected NO PURGE SIGNAL	purge pressure switch (D6) equal to the limit while loaded	0	120 s	automatic	Listen for the regeneration discharge at the silencer during a full dryer cycle and report what you hear with the message; see section 8.18.
W116	Oil level low OIL LEVEL LOW	oil level switch (D7) equal to the limit in state any	0	300 s	automatic	Stop the unit, let the oil settle and read the sight glass. Top up with the specified grade and look for oil traces around the unit; the procedure is part of task 7.3.4.
W117	Separator element pressure drop high SEP ELEMENT DP	discharge line delta above the limit while loaded	1.0 bar (15 psi)	60 s	automatic	Plan task 7.3.6 at the next opportunity and keep an eye on the motor current until then.
X201	Oil temperature very high OIL TEMP V HIGH	oil temperature (T1) above the limit while the unit runs	85 °C (185 °F)	30 s	automatic	Prepare to stop the unit. Reduce the demand, check the cooling air path and the fan, and clear the cause before the controller shuts the unit down.
X202	Discharge pressure high DISCH PRESS HIGH	discharge pressure (P1) above the limit in state any	11.0 bar (160 psi)	10 s	automatic	Open a consumer to lower the pressure and be ready to stop the unit, then follow section 8.11.
X203	Motor current very high MOTOR CURR V HI	motor current (I1) above the limit while loaded, outside the start mask	8.5 A	30 s	automatic	Prepare to stop the unit and have the separator element, the intake valve and the supply voltage checked before the controller trips.
X204	Compressor not unloading NO UNLOAD	line pressure (P2) above the limit while loaded	10.5 bar (152 psi)	30 s	automatic	Stop the unit with the stop key and keep it stopped until the cause has been found; see section 8.16.
S301	Oil temperature shutdown OIL TEMP SHUTDN	oil temperature (T1) above the limit while the unit runs	95 °C (203 °F)	5 s	manual, at the controller	Let the unit cool down, remove the cause and only then reset the message as described in section 3.5.
S302	Discharge pressure shutdown DISCH P SHUTDN	discharge pressure (P1) above the limit in state any	11.5 bar (167 psi)	5 s	manual, at the controller	Check whether the safety valve has blown, find the cause with section 8.11 and reset the message as described in section 3.5.
S303	Motor overload MOTOR OVERLOAD	motor current (I1) above the limit while loaded, outside the start mask	9.5 A	5 s	manual, at the controller	Do not restart the unit repeatedly. Have the drive, the separator element and the supply checked, then reset the message as described in section 3.5.
S304	Motor start failure START FAILURE	load solenoid (D2) equal to the limit and motor current (I1) below the limit in state any	1	20 s	manual, at the controller	Check the emergency stop and the protection relay in the cabinet, then follow section 8.15 before resetting the message.
S305	Line pressure sensor fault LINE P SENSOR	line pressure (P2) below the limit or line pressure (P2) above the limit in state any	-0.5 bar (-7 psi)	10 s	manual, at the controller	Do not bridge the input. Have the transducer and its wiring checked, then reset the message as described in section 3.5.

Code	Message	Trigger	Threshold	Delay	Reset	Action
S306	Oil temperature sensor fault OIL T SENSOR	oil temperature (T1) below the limit or oil temperature (T1) above the limit in state any	-15 °C (5.0 °F)	10 s	manual, at the controller	Have the sensor and its wiring checked before restarting, then reset the message as described in section 3.5.
S307	Emergency stop activated EMERGENCY STOP	External contact Emergency-stop contact on the control panel opens	—	—	manual, at the controller	Find out why the unit was stopped and make sure that nobody is working on it. Release the button by turning it, then reset the message as described in section 3.5.
S308	Motor protection relay tripped MOTOR PROTECTION	External contact Auxiliary contact of the motor protection relay in the control cabinet opens	—	—	manual, at the controller	Leave the relay tripped and call a qualified electrician. Repeated resetting without finding the cause damages the motor.
M401	Oil change due OIL CHANGE DUE	Service counter run_hours reaches the interval of oil_change	—	—	manual, after the service	Arrange task 7.3.4 and reset the service counter afterwards.
M402	Air intake filter service due AIR FILTER DUE	Service counter run_hours reaches the interval of air_filter_service	—	—	manual, after the service	Arrange task 7.3.3 and reset the service counter afterwards.
M403	Oil filter and separator element due OIL FLT SEP DUE	Service counter run_hours reaches the interval of oil_filter_change, separator_element_change	—	—	manual, after the service	Arrange task 7.3.5 and, when it is due, task 7.3.6, then reset the service counter.
M404	Dryer desiccant inspection due DESICCANT DUE	Service counter run_hours reaches the interval of dryer_desiccant_inspection	—	—	manual, after the service	Arrange task 7.3.9 and reset the service counter afterwards.
M405	Condensate drain service due DRAIN SERV DUE	Service counter run_hours reaches the interval of condensate_drain_service	—	—	manual, after the service	Arrange task 7.3.8 and reset the service counter afterwards.
M406	Annual inspection due ANNUAL INSP DUE	Service counter run_hours reaches the interval of annual_inspection	—	—	manual, after the service	Arrange task 7.3.13 with a qualified service technician and reset the service counter afterwards.

The threshold column gives the value the signal is compared with, and the delay beside it how long the condition must last before the message appears. A short delay keeps one disturbed reading from raising a message; a long one lets a slow quantity such as an oil temperature settle. Where the threshold is printed as a parameter number, the limit is programmable: look the parameter up in 4.2²² for its range and its delivered value. A threshold printed as a plain value is fixed in the controller. Messages with no threshold come from a separate input or from a service counter rather than from a measured signal.

3.5 Acknowledging and resetting messages

How a message leaves the display depends on its reset rule, which the list gives for every code.

- **Automatic.** The message disappears as soon as the condition goes away. Most warnings behave this way; nothing has to be acknowledged, but the event is worth recording.
- **Automatic with hysteresis.** The signal has to fall back past the limit by a hysteresis band first, so a value sitting on its limit does not make the message flicker.
- **Manual.** Every shutdown is reset by hand. Remove the cause and let the unit reach a safe state — a hot machine has to cool down — then press RESET. The controller refuses the reset while the condition is still present.
- **Service.** A service message is cleared by the technician who did the work, by resetting the matching counter in the menu. Resetting a counter without doing the work hides the next interval and is never acceptable.

After a reset, and after every stop, the motor is held for the restart delay, parameter P04 (4.2)²³, before it may start again. That keeps it from being restarted while it is still turning down and holds the starts per hour within what the drive is designed for. The pause is normal and is not a fault.

²². Section 4.2, Parameter list

²³. Section 4.2, Parameter list

4 Programmable settings

4.1 Reaching the settings

Press MENU to open the settings menu, step to a parameter with the arrow keys and press ENTER. The value then blinks; change it with the arrow keys and press ENTER again to store it, or press MENU to leave it as it was. A stored value takes effect at once and survives a power failure.

The controller knows two access levels, and the access column of the parameter list says which one a parameter needs.

- **Operator level** is open without a code. It holds the settings of daily running: the regulation pressures, the timers of the load cycle and the ambient warning limits.
- **Service level** is protected by a code held by the maintenance organisation. It holds the limits that protect the machine: the thresholds behind the messages of 3.3²⁴.

Only a trained technician changes a service-level parameter, and only for a written reason. Raising a protection limit does not let the machine work beyond it; it only removes the warning that would have come first. Record every change with its date and both values, so a later fault can be judged against the settings that were active.

4.2 Parameter list

The list gives every programmable parameter with its number, name, accepted range, delivered value and access level. A value outside the range is refused: the display keeps the old value and nothing is stored.

Programmable settings

Parameter	Unit	Min	Default	Max	Description
P01 Cut-in pressure	bar	5 bar (72.5 psi)	8 bar (116 psi)	9.5 bar (138 psi)	Line pressure at which the controller loads the compressor. Lowering it makes the unit run less often but lets the line fall further between two loaded runs. The value must stay below the cut-out pressure by at least the regulation margin.
P02 Cut-out pressure	bar	6 bar (87.0 psi)	10 bar (145 psi)	11 bar (160 psi)	Line pressure at which the controller unloads the compressor. Raising it increases the stored energy in the reservoirs and the power the unit draws at the end of a loaded run. The value must stay above the cut-in pressure by at least the regulation margin. The value must not exceed the maximum working pressure of the unit.
P03 Unloaded run-on time	s	see text			How long the motor keeps running unloaded after the line has reached the cut-out pressure. A longer run-on time answers a new demand without a motor start; a shorter one saves the energy of the unloaded run.
P04 Restart delay	s	see text			Minimum time between a motor stop and the next motor start. It protects the motor against a restart while it is still running down.
P05 Motor start mask time	s	5 s	15 s	60 s	Time after a motor start during which the controller ignores the messages that depend on the load. It covers the current peak and the pressure transients of the start.
P06 Maximum continuous load time	min	5 min	10 min	60 min	How long the unit may stay loaded without reaching the cut-out pressure before the controller raises message W102 (section 3.4). A loaded run that never ends normally points at a demand the unit cannot cover.
P07 Maximum motor starts per hour	/h	1 /h	6 /h	12 /h	How often the motor may start within a rolling hour before the controller raises message W108 (section 3.4). Frequent starts heat the motor windings and wear the contactor.
P08 Low line pressure delay	s	0 s	10 s	60 s	How long the low-pressure switch has to stay closed before the controller raises message W101 (section 3.4). It suppresses the message during a short demand peak.
P09 Oil temperature warning limit	°C	60 °C (140 °F)	75 °C (167 °F)	90 °C (194 °F)	Oil temperature above which the controller raises message W104 (section 3.4). It is the first stage of the oil temperature chain and leaves time to clean the coolers before the unit stops. The value must stay below the oil temperature shutdown warning limit.
P10 Oil temperature shutdown warning limit	°C	70 °C (158 °F)	85 °C (185 °F)	100 °C (212 °F)	Oil temperature above which the controller announces an approaching shutdown with message X201 (section 3.4). The unit keeps running at this stage. The value must stay above the oil temperature warning limit. The value must stay below the oil temperature shutdown limit.
P11 Oil temperature shutdown limit	°C	80 °C (176 °F)	95 °C (203 °F)	110 °C (230 °F)	Oil temperature above which the controller stops the unit with message S301 (section 3.4) and inhibits a restart until the message is reset. The value must stay above the oil temperature shutdown warning limit.
P12 Discharge pressure shutdown warning limit	bar	10 bar (145 psi)	11 bar (160 psi)	11.5 bar (167 psi)	Discharge pressure above which the controller announces an approaching shutdown with message X202 (section 3.4). The value must stay below the discharge pressure shutdown limit.

Parameter	Unit	Min	Default	Max	Description
P13 Discharge pressure shutdown limit	bar	10.5 bar (152 psi)	11.5 bar (167 psi)	12 bar (174 psi)	Discharge pressure above which the controller stops the unit with message S302 (section 3.4). It is the electronic backup of the safety valve and must stay below it. The value must stay above the discharge pressure shutdown warning limit. The value must not exceed the setting of the safety valve.
P14 Motor current warning limit	A	6 A	7.5 A	9 A	Motor current above which the controller raises message W106 (section 3.4) while the unit is loaded. A rising current points at a mechanical load on the airend or at a supply problem. The value must stay below the motor current shutdown warning limit.
P15 Motor current shutdown warning limit	A	7 A	8.5 A	9.5 A	Motor current above which the controller announces an approaching shutdown with message X203 (section 3.4). The value must stay above the motor current warning limit. The value must stay below the motor current shutdown limit.
P16 Motor current shutdown limit	A	8 A	9.5 A	10.5 A	Motor current above which the controller stops the unit with message S303 (section 3.4). It complements, and does not replace, the motor protection relay in the starter. The value must stay above the motor current shutdown warning limit.
P17 Motor current low limit	A	3 A	5 A	6 A	Motor current below which the controller raises message W107 (section 3.4) while the unit is loaded. A loaded run at unloaded current means the airend is not compressing. The value must stay below the motor current warning limit.
P18 Ambient temperature high limit	°C	30 °C (86 °F)	40 °C (104 °F)	50 °C (122 °F)	Cooling-air temperature above which the controller raises message W109 (section 3.4). It mirrors the upper ambient operating limit of the unit, so raising it is only meaningful together with extra ventilation. The value must equal the upper ambient operating limit of the unit.
P19 Ambient temperature low limit	°C	-10 °C (14 °F)	2 °C (36 °F)	10 °C (50 °F)	Cooling-air temperature below which the controller raises message W110 (section 3.4). It mirrors the lower ambient operating limit of the unit, below which condensate in the unit can freeze. The value must equal the lower ambient operating limit of the unit.
P20 Dryer tower period	s	30 s	60 s	300 s	How long one desiccant tower stays in drying service before the controller swaps the towers. A longer period saves purge air but lets the drying tower load up with more moisture.
P21 Dryer changeover timeout	s	see text			How long the controller waits for a tower changeover while the unit delivers air before it raises message W111 (section 3.4). It has to leave room for a complete tower period plus the changeover itself. The value must leave room for at least two complete tower periods.
P22 Dryer purge pressure warning limit	bar	0.2 bar (2.90 psi)	0.5 bar (7.25 psi)	2 bar (29.0 psi)	Purge line pressure above which the controller raises message W103 (section 3.4). A purge line that stays pressurised means the regeneration air is not leaving the dryer freely.
P23 Separator discharge pressure warning limit	bar	0.2 bar (2.90 psi)	0.5 bar (7.25 psi)	2 bar (29.0 psi)	Separator discharge pressure above which the controller raises message W112 (section 3.4) while the unit is loaded. The port should be vented by the open drain during a loaded run.
P24 Reservoir pressure deviation limit	bar	0.2 bar (2.90 psi)	0.5 bar (7.25 psi)	2 bar (29.0 psi)	Difference between the reservoir pressure and the line pressure above which the controller raises message W113 (section 3.4). A lasting difference points at a restriction between the reservoirs and the pneumatic panel.
P25 Separator element pressure drop limit	bar	0.5 bar (7.25 psi)	1 bar (14.5 psi)	2 bar (29.0 psi)	Pressure drop across the separator element above which the controller raises message W117 (section 3.4) while the unit is loaded. A rising drop is the normal ageing signature of the element.

4.3 Rules the controller enforces

Several parameters only make sense in relation to another one, so the controller checks each new value against its neighbours and refuses a combination that would leave the unit without a graded response.

The regulation band keeps a margin. parameter P01 (4.2)²⁵ is delivered at 8.0 bar (116 psi) and parameter P02 (4.2)²⁶ at 10.0 bar (145 psi). The controller refuses a cut-in that comes closer to the cut-out than the margin in the list, and a cut-out that comes closer to the cut-in; without that margin the unit would load and unload continuously. It also refuses a cut-out above the maximum working pressure, 11.0 bar (160 psi), and a discharge pressure shutdown above the safety valve setting, so the controller stops the unit before the valve blows.

Protection limits stay in order. For the oil temperature, the motor current and the discharge pressure the controller holds three limits: a warning, a shutdown warning and a shutdown. The warning must stay below the shutdown warning, and the shutdown warning below the shutdown. A value that breaks the order is refused, so the unit can never reach a shutdown unannounced.

Ambient warnings follow the operating limits. parameter P18 (4.2)²⁷ and parameter P19 (4.2)²⁸ are tied to the operating limits given in 5.1²⁹. They report that the machine room has left the range the unit was designed for; moving them does not move that range.

24. Section 3.3, Message types

25. Section 4.2, Parameter list

26. Section 4.2, Parameter list

27. Section 4.2, Parameter list

28. Section 4.2, Parameter list

29. Section 5.1, Site

Dryer timers keep their ratio. parameter P21 (4.2)³⁰ has to stay well above parameter P20 (4.2)³¹, because the timeout must allow at least one complete changeover before it reports a dryer that has stopped working.

30. Section 4.2, Parameter list
31. Section 4.2, Parameter list

5 Installation and reference conditions

This chapter covers the installation of the unit and the conditions its ratings refer to. Read [1.1³²](#) before you begin, and have the work done by people who are allowed to do it.

5.1 Site

Install the unit indoors, on a level floor that carries its mass of 320 kg (705 lb), in a room that stays clean, dry and frost-free. It measures 1200 mm (47.24 in) in length, 800 mm (31.5 in) in width and 1350 mm (53.15 in) in height. Leave room on every side to open the panels and pull the coolers; a clear space equal to the width of the unit in front of the cooling-air inlet and behind the outlet is the practical minimum.

The room must supply the cooling air the unit needs and get rid of the heat again. Do not let warm outlet air return to the inlet. Draw the intake air from the cleanest place available: dust shortens the life of the intake filter, the separator element and the desiccant alike.

Keep the cooling air between 2 °C (36 °F) and 40 °C (104 °F). [2.4³³](#) explains why. The relative humidity of the room must stay below 90 %, and the site must lie no higher than 1000 m (3281 ft) above sea level; thinner air carries away less heat and reduces the delivered flow. When the unit is stored rather than installed, keep it between -20 °C (-4.0 °F) and 60 °C (140 °F).

5.2 Reference conditions

Every rating in this book, and every band the controller compares a reading with, refers to one agreed set of conditions. They are an ambient temperature of 20 °C (68 °F), cooling air at 20 °C (68 °F), an absolute inlet pressure of 1.0 bar (15 psi) and a relative humidity of 0 %.

At reference conditions therefore means two things. A rating such as the free air delivery or the pressure dew point is the value the unit reaches when the site matches that set; a warmer, damper or higher site gives less. And the normal bands in [9.3³⁴](#) were recorded on a unit running at those conditions, so a reading that sits outside its band on a hot day is not automatically a fault. Judge it against the ambient temperature of the moment, and note the conditions whenever you log a measurement. The ratings themselves are collected in [9.1³⁵](#).

5.3 Electrical connection

Connect the unit to a three-phase supply of 400 V at 50 Hz, through an external disconnecting device that can be locked in the open position. The supply voltage may deviate by 10 % from its nominal value; a larger deviation, or an unbalanced supply, overloads the motor.

Protect the circuit with a fuse of 16 A, sized for the rated motor current of 7.0 A and for the peak the motor draws at each start. Use a cable cross-section that suits the fuse and the length of the run, and earth the unit before you connect anything else.

Check the direction of rotation on the first start. A screw compression element that turns the wrong way delivers no air and is damaged within seconds: start the motor briefly, watch the arrow on the drive and swap two phases if it turns wrongly. [1.3³⁶](#) applies throughout.

5.4 Air connection

Connect the delivery pipe to the outlet of the reservoir isolation valve through a flexible element, so that no pipe force reaches the unit. Size the pipe for the flow the installation takes and lay it with a fall towards a drain point. Close the isolation valve to separate the unit from the distribution line during service work, and leave it open in normal operation: a closed valve makes the unit cycle against its own reservoirs.

Lead the condensate from the automatic drain to a collecting vessel or an oil-water separator through a pipe that falls all the way; it carries oil and must not go into the drain untreated. Never connect the drain line to a line that can be pressurised from elsewhere.

Leave the outlet of the purge silencer free. It discharges to atmosphere in short bursts and must never be piped into a closed system or throttled; back pressure in the purge line stops the off-line tower from regenerating. [2.2³⁷](#) shows where each of these connections sits in the air path.

5.5 Commissioning

Work through this list before the first start, and again after any work that opened the pressure envelope or the electrical box.

- All transport brackets removed, all panels back in place and all guards fitted.

32. Section 1.1, General rules

33. Section 2.4, Cooling

34. Section 9.3, Normal bands

35. Section 9.1, Technical data

36. Section 1.3, Electrical supply

37. Section 2.2, Air flow

- Oil level in the separator vessel above the minimum mark, with the unit stopped and vented.
- Electrical connection, earth and direction of rotation checked as described above.
- Reservoir isolation valve open, condensate line connected and falling, purge silencer outlet free.
- Cooling-air inlet and outlet unobstructed, room temperature inside the window given above.
- Cut-in and cut-out pressures checked against the installation they have to serve.

Then start the unit as described in [6.1](#)³⁸ and let it run through several complete load cycles while you watch the display. Confirm that the unit unloads at the cut-out pressure, that it loads again at the cut-in pressure, that the condensate drain discharges and that the dryer changes over. Finally run [task 7.3.1](#)³⁹ once, so that the first log entry records the unit when it was known to be sound.

38. Section 6.1, Starting the unit

39. Section 7.3.1, Daily checks

6 Operation

6.1 Starting the unit

Work through these checks before the first start of a shift and after every service.

1. Read the oil level while the unit is stopped and vented.
Top up before starting, never during operation.
2. Check that no message is active. A shutdown has to be cleared as described in 3.5⁴⁰ before the unit will start at all.
3. Check that the reservoir isolation valve is open and the condensate drain lines run free.
4. Check that the cooling air inlet and outlet are clear and all guards are fitted.
5. Check that the emergency-stop button is released.

Press START. The controller closes the intake valve, starts the motor and lets the unit run unloaded while the oil circuit fills and the drive settles. It then compares line pressure (P2) with the cut-in setting: the unit goes to load only while the line pressure is **below** 8.0 bar (116 psi), and otherwise stays unloaded until the consumers have drawn the pressure down. On the reference machine, loading begins at 8.05 bar (117 psi).

For a few seconds after the motor starts, motor current (I1) is higher than its running value and can approach the rated current of the motor, 7.0 A. The controller therefore ignores the motor current messages for the start mask time, 15 s, so the peak raises no warning.

After every stop, and after a message has been reset, the controller holds the motor for the restart delay of 30 s before it will start again. A START command given inside that window is remembered and carried out when the delay has run out.

6.2 Stopping the unit

Press STOP for a normal stop. The controller does not cut the motor at once; it unloads the unit first. The intake valve closes, the blow-down valve vents the oil separator vessel so that discharge pressure (P1) falls to near zero, and the motor keeps turning unloaded for the run-on time before it stops. Running on lets the oil drain back and leaves the compression element unpressurised, so the next start is made against a closed intake.

Use the emergency-stop button only in an emergency. It stops the motor immediately, without the unloading and run-on sequence, and raises a shutdown message. Release it by turning it, then reset the message; the unit will not start before both are done.

A stopped unit is not a vented unit. The blow-down valve vents the separator vessel only; the reservoirs, the dryer and the distribution system stay at line pressure (P2). Before any work on the air side, isolate the unit, close the reservoir isolation valve, vent the section you will open and check that the panel reads no pressure.

6.3 The normal load cycle

In steady operation the unit repeats one cycle. The figures below describe the machine at its reference operating point — what a healthy unit looks like — and the bands they come from are in 9.3⁴¹.

Cut-in. The consumers draw the line pressure down to the cut-in setting, 8.05 bar (117 psi) on the reference machine. The controller energises the load solenoid and opens the intake valve.

Loaded run. The unit delivers. line pressure (P2) rises at about 1.1 bar/min (16 psi/min) and discharge pressure (P1) follows it, a little higher because of the separator element. A loaded run lasts about 109 s, almost always between 99 s and 129 s. During the run intake valve closed (D1) reads zero, load solenoid (D2) reads one, separator discharge pressure (P3) falls to near zero because the separator discharges into the delivery, and motor current (I1) sits at its loaded value.

Cut-out. At the cut-out setting, 10.03 bar (145.5 psi) on the reference machine, the controller unloads. The intake valve closes, the blow-down valve vents the separator vessel and discharge pressure (P1) drops to near zero, while separator discharge pressure (P3) rises to the line pressure again.

Run-on. The motor keeps turning unloaded for 420 s, so that a short new demand can be answered without a restart. The reference machine runs on for about 407 s.

Off. If no new demand arrives the motor stops. The off phase lasts about 1329 s, during which motor current (I1) reads zero and the line pressure decays slowly through the normal leakage of the system, at about 0.07 bar/min (1 psi/min). Together that is a load rate of about 1.97 /h.

6.4 Checks during operation

Walk past the unit once a shift and read the panel. Judge every value against the machine state it was read in, because the normal band of a signal differs between loaded, unloaded and off; the bands are in 9.3⁴².

- oil temperature (T1) settles inside its band a few minutes after a cold start. A reading that keeps climbing through a loaded run, or never leaves the bottom of the band, is worth following up.
- motor current (I1) has its own band per state and is highest while the unit is loaded. Compare it with the band of the state the unit is actually in.
- dryer purge pressure (P4) reads near zero while the unit delivers, apart from a short rise at each tower changeover.
- dryer tower in service (D3) changes over while the unit delivers, so the towers share the work.
- The condensate drain discharges briefly and regularly. A drain that blows continuously and one that never discharges are both faults.
- Note the code and the reading of any message before you acknowledge it; the reading is what separates the possible causes later.

⁴⁰. Section 3.5, Acknowledging and resetting messages

⁴¹. Section 9.3, Normal bands

⁴². Section 9.3, Normal bands

Four departures from that pattern matter more than the exact values, because each points at a group of causes.

- **More frequent cycles.** The unit loads more often over the same shift and the loaded runs get shorter: see 8.4⁴³.
- **Longer loaded runs, or no cut-out.** The unit stays loaded and the line pressure no longer reaches the cut-out setting: see 8.5⁴⁴.
- **Faster pressure decay.** The off phase grows shorter because the pressure falls away quicker than the reference figure above.
- **Purge pressure that stays up.** dryer purge pressure (P4) no longer returns to near zero between changeovers, so air is leaving through the regeneration path: see 8.6⁴⁵.

6.5 Dryer operation

The twin-tower dryer works without heat. One tower dries the delivered air while the other is regenerated by a small flow of dried air, which expands to atmosphere through the purge silencer and carries the collected moisture with it. The towers then change over and swap roles.

Changeover is tied to delivery, not to the clock: the towers change over only while the unit is loaded, and the tower period, 60 s, counts loaded time only. A unit that stands unloaded for a long time therefore keeps the same tower in service, which is correct, because a tower that dries no air needs no regenerating. After a cut-in the first changeover follows within about 60 s.

If the controller sees no changeover for 180 s of loaded running, it reports the dryer. The timeout is generous on purpose, so that a short loaded run followed by a long off phase never reports a dryer that is working.

A short, sharp discharge at the purge silencer at every changeover is normal, and so is the noise it makes. A discharge that never stops is not: compressed air is then lost continuously through the regeneration path, and it shows up as a purge pressure that no longer falls back.

With both towers working and the desiccant in good condition the unit delivers air at a pressure dew point of -40 °C (-40 °F) at the reference conditions. Moisture at the consumers, a risen dew point or water in a filter bowl point at the dryer rather than the compressor.

43. Section 8.4, Compressor starts and loads too often

44. Section 8.5, Compressor stays loaded and does not reach cut-out

45. Section 8.6, Dryer purge pressure high, air escaping at the purge silencer

7 Maintenance

7.1 Safety during maintenance

Only trained personnel may service the CAU-7 Compressed-Air Unit. Read chapter 1⁴⁶ before you start, and carry out the four steps below, in this order, before every task in this chapter. They apply to the short checks as well as to the long overhauls; most of the accidents a unit of this kind causes happen during a job that was thought too small to prepare for.

Stop the unit and wait for it to come to rest. Stop it from the controller and let the motor run down. A unit that is merely unloaded is not stopped: the CTRL-7 loads the motor again on its own as soon as the line pressure falls to the cut-in setting, and it gives no warning before it does so.

Isolate the unit electrically and secure it against restarting. Lock the isolator open and keep the key with the person doing the work. A warning label hung on an unlocked isolator is not a lock-out; see 1.3⁴⁷.

Vent the unit. The blow-down valve empties the separator vessel when the unit stops, but the reservoirs, the dryer towers and the pipework stay under pressure until you close the isolation valve at the pneumatic panel and open the vent. Read discharge pressure (P1) and reservoir pressure (P5) on the controller, confirm that both stand at zero and confirm it a second time on a test gauge at the panel before you open any joint. Stored compressed air gives no sign of itself, and a vessel that still holds pressure will throw a cover bolt or a fitting across the room; see 1.2⁴⁸.

Let the unit cool down. The airend, the oil charge, the separator vessel, the oil cooler and the aftercooler stay hot long after the motor has stopped. Hot oil under residual pressure sprays out of a joint that is opened too early; see 1.4⁴⁹.

Never adjust, block or paint the safety valve, and never defeat a guard or a door interlock to reach a part more comfortably. Work on the electrical cabinet, on the safety valve and on the pressure vessels is reserved for qualified personnel. When

the work is finished, refit every guard, take every tool out of the enclosure, close the drain and vent valves that you opened, and only then restore the supply and unlock the isolator.

7.2 Maintenance schedule

The schedule below gives the interval of every task. An interval stated in both running hours and elapsed months falls due at whichever comes first: a unit that stands idle ages through condensate and through oxidation of the oil just as a busy unit ages through running hours, so neither figure on its own is enough. The controller counts the running hours for you; the calendar months are yours to keep.

The daily and the weekly checks are operator work and need no tool beyond a torch. Every task from the air intake filter service onwards is service work, and each of them assumes that the unit has been stopped, isolated, vented and allowed to cool as 7.1⁵⁰ describes.

The tasks that the controller can count carry a service message. The message appears once the counter reaches the interval, it does not stop the unit, and it stays on the display until the work has been carried out and the counter has been reset. 3.3⁵¹ explains how a service message differs from a warning: a warning reports a reading that has left its band now, a service message reports work that has become due. The remaining tasks — the daily and weekly checks, the cooler cleaning, the purge valve and silencer service, the safety valve test and the drive coupling inspection — are not counted by the controller and have to be planned from the operating log.

Site conditions shorten intervals; they never lengthen them. A dusty room, a warm room, a high duty cycle or a long spell of continuous load wears the intake filter, the oil charge and the desiccant faster than the figures below assume. Shorten the interval when the unit works in such a place, rather than waiting for the message and then finding a filter that has been choked for months.

Maintenance schedule

Task	Interval	Consumables
Daily checks	daily	none
Weekly checks	weekly	none
Air intake filter service	2000 h / 12 months	1 Air intake filter element (CAU7-AF-01)
Oil change	4000 h / 12 months	8 L (2.11 US gal) Synthetic screw compressor oil (CAU7-OIL-46)
Oil filter change	4000 h	1 Oil filter element (CAU7-OF-01)
Separator element change	8000 h / 24 months	1 Oil separator element (CAU7-SE-01)
Oil cooler and aftercooler cleaning	2000 h / 6 months	1 Cooler cleaning kit (CAU7-CK-01)
Condensate drain service	2000 h	1 Condensate drain valve service kit (CAU7-DV-01)
Dryer desiccant inspection	8000 h / replacement at 16,000 h	2 Desiccant charge for one tower (CAU7-DC-35)
Purge valve and silencer service	4000 h	1 Dryer purge valve service kit (CAU7-PV-01), 1 Purge silencer (CAU7-PS-01)
Safety valve test	12 months	none
Drive coupling inspection	8000 h	1 Coupling element (CAU7-CE-01)
Annual inspection	12 months	none

46. Chapter 1, Safety precautions

47. Section 1.2, Pressure

48. Section 1.2, Pressure

49. Chapter 1, Safety precautions

50. Section 1.2, Pressure

51. Section 3.3, Message types

order in which they are carried out and the checks that close it. Read the whole procedure before you begin and lay out the parts first: a procedure that stops halfway for a missing seal leaves the unit open, unusable and, if the enclosure has been left standing open overnight, dirty inside as well.

Use only the consumables listed for the task. The oil grade is part of the design of the unit; an oil of another grade changes the viscosity, the separation behaviour and the running temperature that the controller has been set up to expect, and the effect shows up weeks later as oil carry-over or as a warning nobody can explain. Fit the new seals and gaskets that a procedure names, even when the old ones still look sound.

7.3.1 Daily checks

Interval daily · 10 min · tools: Torch

WARNING

Keep clear of hot surfaces and of the cooling fan while the unit is running; see [section 1.4](#).

Do not open any part of the air or oil circuit for these checks; see [section 1.2](#).

1. Read the controller display and note every pending or unacknowledged message.
2. Stop the unit, wait until it has vented and read the oil level through the sight glass.
3. Check that oil level switch (D7) reports oil above the minimum on the controller.
4. Watch the condensate drain over one discharge and confirm that it opens and closes again.
5. Walk around the unit and listen for escaping air at the purge silencer and at the pipework.
6. Compare line pressure (P2) and discharge pressure (P1) with the values in [section 9.3](#).
7. Note the readings in the operating log so that a slow drift becomes visible.

After the service, check that:

- No unacknowledged message remains on the controller.
- The oil level stands between the marks on the sight glass with the unit stopped.
- The condensate drain is dry and quiet between its discharges.

Signals to watch after the service: [P3](#) Separator discharge pressure, [P4](#) Dryer purge pressure, [P2](#) Line pressure, [D5](#) Low-pressure switch, [D7](#) Oil level switch, [T1](#) Oil temperature.

7.3.2 Weekly checks

Interval weekly · 20 min · tools: Torch, Soft brush

WARNING

Carry out the checks with the unit stopped, isolated and vented wherever a guard has to be opened; see [section 7.1](#).

Keep clear of hot surfaces until the unit has cooled down; see [section 1.4](#).

1. Repeat the daily checks of [task 7.3.1](#) before you start.
2. Brush the dust off the cooler matrix and the ventilation grilles with the unit stopped.
3. Check the intake filter housing for dust marks and for a loose or damaged seal.
4. Follow one complete load cycle and confirm that the unit loads, delivers and unloads again.
5. Confirm that the dryer changes tower during the cycle and that the regeneration discharge is audible.
6. Compare the loaded run time and the number of load cycles with the figures in [section 9.3](#).

7. Inspect the flexible hoses and the pipe supports for chafing, oil traces and loose clamps.
8. Record the run hours and the readings in the operating log.

After the service, check that:

- The unit reaches the cut-out pressure and unloads within the expected run time.
- The dryer changes tower and regeneration is audible at the silencer.
- No oil trace and no loose connection remain on the unit.

Signals to watch after the service: [I1](#) Motor current, [T1](#) Oil temperature, [T2](#) Ambient temperature, [P4](#) Dryer purge pressure, [D5](#) Low-pressure switch, [D3](#) Dryer tower in service, [D6](#) Purge pressure switch.

7.3.3 Air intake filter service

Interval 2000 h / 12 months · 30 min · service message **M402** · tools: Socket set, Soft brush, Torch

WARNING

Stop the unit, isolate it electrically and secure it against restarting; see [section 1.3](#).

Let the unit vent before opening the intake housing; see [section 1.2](#).

1. Stop the unit, isolate it and wait until the display reports no pressure.
2. Open the intake filter housing and take out the used element.
3. Wipe the inside of the housing clean, working from the clean side outwards.
4. Inspect the housing seal and the intake duct for cracks and for loose clamps.
5. Fit the new element and close the housing so that the seal is evenly compressed.
6. Restore the supply, start the unit and follow one load cycle.
7. Reset the service counter for this task at the controller.

After the service, check that:

- The motor current under load is back inside the band of [section 9.3](#).
- The housing is airtight and no air is drawn in past the seal.
- The service message no longer stands on the controller.

Signals to watch after the service: [I1](#) Motor current, [P1](#) Discharge pressure.

7.3.4 Oil change

Interval 4000 h / 12 months · 60 min · service message **M401** · tools: Socket set, Drain hose, Collecting vessel, Funnel

WARNING

Stop the unit, isolate it electrically, secure it against restarting and vent the separator vessel; see [section 7.1](#).

Hot oil causes burns; let the unit cool before draining; see [section 1.4](#).

Collect the used oil and hand it over to an authorised disposal service.

1. Run the unit until it is warm, then stop it, isolate it and vent the separator vessel.

2. Check that discharge pressure (P1) reads zero on the controller before you open anything.
3. Connect the drain hose and drain the used oil into the collecting vessel.
4. Drain the oil cooler and the oil lines as well, so that no old oil stays in the circuit.
5. Close the drain, refill through the filler port with the specified grade and check the sight glass.
6. Refit the filler plug with a new sealing ring and wipe the unit dry.
7. Restore the supply, start the unit and let it run through a full load cycle.
8. Stop the unit again, let the oil settle and correct the level if needed.
9. Reset the service counter for this task at the controller.

After the service, check that:

- The oil level stands between the marks on the sight glass with the unit stopped and vented.
- No leak shows at the drain, at the filler port or at the cooler after the test run.
- The oil temperature settles inside the band of [section 9.3](#).
- The service message no longer stands on the controller.

Signals to watch after the service: [T1](#) Oil temperature, [I1](#) Motor current, [D7](#) Oil level switch, [P1](#) Discharge pressure.

7.3.5 Oil filter change

Interval 4000 h · 30 min · service message **M403** · tools: Filter wrench, Collecting vessel, Lint-free cloth

WARNING

Stop the unit, isolate it electrically and vent the separator vessel before opening the oil circuit; see [section 7.1](#).

Let the oil cool before touching the filter head; see [section 1.4](#).

1. Stop the unit, isolate it, vent it and place the collecting vessel under the filter head.
2. Unscrew the used filter element and let the oil in it drain into the vessel.
3. Clean the sealing face on the filter head with a lint-free cloth.
4. Oil the gasket of the new element with clean compressor oil and screw it on by hand.
5. Tighten the element a further part turn as marked on it, without a wrench.
6. Restore the supply, start the unit and check the filter head for leaks under pressure.
7. Correct the oil level after the test run and reset the service counter at the controller.

After the service, check that:

- No oil escapes at the filter head under full working pressure.
- The oil level stands between the marks after the test run.
- The oil temperature settles inside the band of [section 9.3](#).

Signals to watch after the service: [T1](#) Oil temperature, [I1](#) Motor current.

7.3.6 Separator element change

Interval 8000 h / 24 months · 90 min · service message **M403** · tools: Socket set, Torque wrench, Lint-free cloth, Lifting aid

WARNING

Stop the unit, isolate it electrically, secure it against restarting and vent the separator vessel completely; see [section 7.1](#).

The separator vessel is a pressure vessel; never open it while it holds pressure; see [section 1.2](#).

1. Stop the unit, isolate it and vent it, then confirm that discharge pressure (P1) reads zero.
2. Disconnect the scavenge line and the pipework from the separator cover and mark their positions.
3. Undo the cover bolts evenly and lift the cover off with the lifting aid.
4. Take out the used element and clean the sealing faces in the vessel and on the cover.
5. Clear the scavenge line and its orifice, and renew the strainer in it if one is fitted.
6. Fit the new element with new gaskets and make sure that its earthing strap is in place.
7. Refit the cover and tighten the bolts crosswise to the torque given on the cover plate.
8. Reconnect the scavenge line and the pipework, then top up the oil level.
9. Restore the supply, start the unit and let it reach working pressure while you check for leaks.
10. Reset the service counter for this task at the controller.

After the service, check that:

- The pressure drop across the element under load is back inside the band of [section 9.3](#).
- No air and no oil escape at the cover or at the scavenge line under full working pressure.
- The motor current under load has fallen back to its normal value.
- The service message no longer stands on the controller.

Signals to watch after the service: [P1](#) Discharge pressure, [I1](#) Motor current, [P2](#) Line pressure, [T1](#) Oil temperature, [D7](#) Oil level switch.

7.3.7 Oil cooler and aftercooler cleaning

Interval 2000 h / 6 months · 45 min · tools: Socket set, Soft brush, Compressed-air gun, Eye protection

WARNING

Stop the unit, isolate it electrically and secure it against restarting before removing any guard; see [section 1.3](#).

Let the coolers cool down before touching them; see [section 1.4](#).

Wear eye protection while blowing out the matrix and keep other people away.

1. Stop the unit, isolate it and let the coolers cool down.
2. Remove the guard in front of the cooling fan and check the fan blades for damage and deposits.
3. Blow the cooler matrix out against the normal air flow direction, working from the clean side.
4. Brush the remaining deposits off the fins carefully, without bending them.
5. Clean the ventilation grilles of the enclosure and the air path in front of the unit.
6. Refit the guard, restore the supply and start the unit.

7. Follow one full load cycle and watch the oil temperature settle.

After the service, check that:

- The oil temperature under load is back inside the band of [section 9.3](#).
- The cooling fan runs freely and the guard is properly fastened.
- No deposit blocks the ventilation grilles of the enclosure.

Signals to watch after the service: [T1](#) Oil temperature, [T2](#) Ambient temperature, [I1](#) Motor current.

7.3.8 Condensate drain service

Interval 2000 h · 30 min · service message **M405** · tools: Spanner set, Collecting vessel, Lint-free cloth

WARNING

Stop the unit, isolate it electrically and vent the separator before opening the drain; see [section 7.1](#).

Condensate contains oil; collect it and hand it over to an authorised disposal service.

1. Stop the unit, isolate it, vent it and place the collecting vessel under the drain.
2. Close the isolation valve upstream of the drain and open the manual test valve to relieve it.
3. Dismantle the drain valve and clean the strainer, the seat and the discharge port.
4. Renew the seals, the diaphragm and the strainer from the service kit.
5. Reassemble the drain, open the isolation valve and clear the discharge line to the collector.
6. Restore the supply, start the unit and watch the drain over several discharges.
7. Reset the service counter for this task at the controller.

After the service, check that:

- The drain opens and closes cleanly and does not leak air between discharges.
- The separator discharge pressure under load is back inside the band of [section 9.3](#).
- No condensate stands in the separator after a full cycle.

Signals to watch after the service: [P3](#) Separator discharge pressure, [D3](#) Dryer tower in service, [P4](#) Dryer purge pressure.

7.3.9 Dryer desiccant inspection

Interval 8000 h / replacement at 16,000 h · 60 min · service message **M404** · tools: Spanner set, Dust mask, Eye protection, Funnel, Collecting vessel

WARNING

Stop the unit, isolate it electrically and vent both dryer towers before opening them; see [section 7.1](#).

Desiccant dust irritates the eyes and the airways; wear a dust mask and eye protection.

1. Stop the unit, isolate it and vent both towers through the purge line.
2. Open the fill port of the first tower and take a sample of the desiccant.
3. Judge the sample for dust, for broken beads and for oil contamination.

4. Repeat the inspection on the second tower and compare the two samples.
5. Replace the charge in both towers when the beads are dusty, oily or broken down.
6. Refit the fill ports with new seals and clear any dust from the sieve plates.
7. Restore the supply, start the unit and let the dryer run through several changeovers.
8. Reset the service counter for this task at the controller.

After the service, check that:

- Both towers change over at the expected interval and regeneration is audible each time.
- The delivered air shows no visible moisture at the test point downstream of the dryer.
- The purge pressure between changeovers stays inside the band of [section 9.3](#).

Signals to watch after the service: [D3](#) Dryer tower in service, [P4](#) Dryer purge pressure, [T2](#) Ambient temperature, [D6](#) Purge pressure switch, [P3](#) Separator discharge pressure.

7.3.10 Purge valve and silencer service

Interval 4000 h · 60 min · tools: Spanner set, Hex key set, Lint-free cloth, Ear protection

WARNING

Stop the unit, isolate it electrically and vent the dryer before working on the purge line; see [section 7.1](#).

Wear ear protection when you later listen to the regeneration discharge; see [section 1.1](#).

1. Stop the unit, isolate it and vent the dryer towers and the purge line.
2. Remove the purge silencer and check it for blockage and for mechanical damage.
3. Dismantle the purge valve, clean the seat and inspect it for scoring.
4. Renew the seals and the valve seat from the service kit and reassemble the valve.
5. Fit the new silencer and make sure that its outlet is unobstructed.
6. Check the changeover valves and the control tubing for leaks and for loose fittings.
7. Restore the supply, start the unit and follow two complete dryer changeovers.

After the service, check that:

- The purge pressure under load is back inside the band of [section 9.3](#).
- The regeneration discharge stops cleanly at the end of each purge phase.
- No permanent discharge is audible at the silencer while the unit is delivering.

Signals to watch after the service: [P4](#) Dryer purge pressure, [T1](#) Oil temperature, [I1](#) Motor current, [D5](#) Low-pressure switch, [D6](#) Purge pressure switch, [D3](#) Dryer tower in service.

7.3.11 Safety valve test

Interval 12 months · 20 min · tools: Ear protection, Eye protection, Test certificate form

WARNING

Only a qualified technician may test the safety valve; see [section 7.1](#).

The valve discharges loudly and suddenly; wear ear and eye protection and keep other people away; see [section 1.2](#).

Never block, adjust or paint the safety valve.

1. Read the setting stamped on the safety valve and compare it with the technical data of the unit.
2. Check the valve, its connection and its discharge path for corrosion and for obstruction.
3. Raise the pressure in a controlled way or lift the valve by its test lever, as its type requires.
4. Confirm that the valve opens at its setting and reseats cleanly when the pressure falls.
5. Look for leakage at the seat after the test with the unit at working pressure.
6. Replace the valve if it leaks, sticks or opens away from its setting.
7. Record the result and the date on the test certificate kept with the unit.

After the service, check that:

- The valve reseats without leakage at working pressure.
- The discharge path is clear and points away from the operating position.
- The test result is recorded in the documents that stay with the unit.

Signals to watch after the service: [P1](#) Discharge pressure, [P2](#) Line pressure, [I1](#) Motor current, [T1](#) Oil temperature.

7.3.12 Drive coupling inspection

Interval 8000 h · 45 min · tools: Socket set, Feeler gauge, Straight edge, Torque wrench

WARNING

Stop the unit, isolate it electrically and secure it against restarting before removing the coupling guard; see [section 1.3](#).

Never turn the drive by hand while the supply is connected; see [section 7.1](#).

1. Stop the unit, isolate it, secure it against restarting and remove the coupling guard.
2. Turn the drive by hand and listen for roughness in the motor and airend bearings.
3. Inspect the flexible coupling element for cracks, for rubber dust and for permanent set.
4. Replace the coupling element when it shows cracks, dust or excessive play.
5. Check the alignment of motor and airend with the straight edge and the feeler gauge.
6. Tighten the motor and airend feet to the torque given in the technical data.
7. Refit the guard, restore the supply and start the unit while you listen for vibration.

After the service, check that:

- The guard is refitted and no tool is left inside the enclosure.

- The unit starts without unusual noise and the current peak at start is as expected.
- No new vibration is felt on the frame under load.

Signals to watch after the service: [I1](#) Motor current, [T1](#) Oil temperature.

7.3.13 Annual inspection

Interval 12 months · 180 min · service message **M406** · tools: Socket set, Multimeter, Torque wrench, Torch, Inspection record form

WARNING

Only qualified personnel may carry out this inspection; see [section 7.1](#).

Isolate the unit electrically and vent it before opening the cabinet or the air circuit; see [section 1.3](#).

1. Review the operating log and the message history since the last inspection.
2. Carry out the checks of task 7.3.2 and record their results.
3. Confirm that task 7.3.11 has been carried out and is documented.
4. Check the electrical cabinet for loose terminals, heat marks and damaged insulation.
5. Measure the supply voltage on all three phases and confirm that it is balanced.
6. Compare the controller settings with the values recorded at commissioning.
7. Check the pressure vessels, their documents and their corrosion protection.
8. Check the emergency stop, the motor protection relay and the door interlocks for correct operation.
9. Compare the readings of every sensor with the bands in [section 9.3](#).
10. Record the inspection and reset the service counter at the controller.

After the service, check that:

- Every safety device has been operated and has responded correctly.
- The controller settings match the commissioning record or the deviations are documented.
- The inspection record is complete and is kept with the unit.
- The service message no longer stands on the controller.

Signals to watch after the service: [I1](#) Motor current, [T1](#) Oil temperature, [D2](#) Load solenoid, [D1](#) Intake valve closed.

7.3.14 Recording the work and resetting the service message

Record every service in the operating log: the date, the running hours, the task, the parts that were fitted and the readings taken before and after the work. The log is what turns a row of separate repairs into a history. A slow drift in oil temperature (T1), in the loaded run time or in the number of load cycles per hour is invisible from one visit to the next and obvious across a year of entries, and most of the conditions in [chapter 8⁵²](#) are found that way long before a message appears.

Reset the service counter at the controller once the work is complete and the post-service checks have passed, and not before. Resetting a counter clears the service message and starts the next interval from the current running hours; [3.5⁵³](#)

gives the key sequence. A counter that is cleared without the work being done hides the task until the next interval falls due, and the unit then runs with a choked filter, a tired oil charge or a spent desiccant charge for twice as long as was intended.

Close every visit by running the unit through one complete load cycle with all guards refitted, and compare the readings with the figures in 9.3⁵⁴ before you hand the unit back to the operator.

52. Chapter 8, Problem solving

53. Section 3.5, Acknowledging and resetting messages

54. Section 9.3, Normal bands

8 Problem solving

8.1 How to use this chapter

Start from what the unit shows, not from the part you suspect. Each section of this chapter is one condition: the symptom as the operator meets it, the controller messages that usually come with it, the readings that carry the evidence, and a table of the causes that produce it.

Read a table in four columns. The **cause** names the thing that is wrong. The **likelihood** ranks it against the other causes of the same condition, and the rows are already ordered, so work down the list rather than across it. The **signals** column says which readings move, in which direction and in which machine state; that is the column which tells you whether the unit in front of you is really showing this cause or only this condition. The **remedy** says what to do once the cause is confirmed. Each cause also carries its own checks, and the checks come before the spanner: they confirm or rule out a cause without changing a part.

Before you call for service, write down what the controller shows. Note the machine state, the code and the text of every pending message, the running hours, and the readings for line pressure (P2), discharge pressure (P1), oil temperature (T1), motor current (I1), dryer purge pressure (P4) and separator discharge pressure (P3) in each state the unit passes through. Add how long a loaded run lasts and how often the unit loads in an hour. Those figures, compared against 9.3⁵⁵, are worth more than any description of the noise the unit makes.

Not every cause is a fault. Some of the causes listed here are benign: the unit is healthy and is reporting what the plant or the room is doing to it. Air demand above the rated delivery and an ambient temperature outside the operating range both behave exactly like a defect for as long as they last, and both disappear on their own. Recognise them, record them and change the installation or the duty; there is nothing on the machine to repair.

Several conditions share the same causes, which is why the checks decide and the symptom alone does not. Two examples are worth reading before you use the tables.

A leak in the distribution network is listed under 8.3⁵⁶ and again under 8.5⁵⁷. It is the same fault in both places; only the stage differs. Early on, the unit still reaches its cut-out setting and the pressure simply falls back faster than it should during the idle phase, so the unit loads more often. Later, when the loss has grown to match what the unit can deliver, the pressure settles on a plateau and the unit stays loaded. The check is the same at either stage: stop the unit, close the isolation valve at the pneumatic panel and watch whether the reservoir pressure still falls. If it holds, the loss is in the plant behind the valve. The entry to follow is Leak in the distribution network (8.3)⁵⁸.

A purge valve that no longer seats is listed under 8.5⁵⁹ as well, and on the display the two faults look identical: the unit delivers without pause and never reaches cut-out. The pressure-hold test separates them, because this loss is inside the unit and the pressure keeps falling with the panel valve closed. So does the purge line itself: dryer purge pressure (P4) stays raised between tower changeovers instead of falling back, and a steady discharge is audible at the silencer. The reported condition is the same; the check tells you which cause to work on, and here that cause is Dryer purge valve not seating (8.3)⁶⁰.

8.2 Symptoms, causes and remedies

The conditions below are arranged from the pressure faults an operator meets first, through the oil, cooling, separation and drying faults, to the electrical and sensor faults that stop the unit outright.

55. Section 9.3, Normal bands

56. Section 8.3, Line pressure below setpoint

57. Section 8.5, Compressor stays loaded and does not reach cut-out

58. Section 8.3, Line pressure below setpoint

59. Section 8.5, Compressor stays loaded and does not reach cut-out

60. Section 8.3, Line pressure below setpoint

Symptoms, causes and remedies

Condition	Fault id		Subsystem
Possible cause	What to check	Remedy	See also
8.3 Line pressure below setpoint low_line_pressure — The display shows a line pressure below the cut-in setting and message W101 (section 3.4) appears once the delay has run out. Consumers lose pressure although the unit is running, and the low-pressure switch may close. Loaded runs last far longer than the reference cycle in section 6.3.			
low_line_pressure	downstream_air_leak		distribution
Leak in the distribution network (common) Air escapes from the pipework, hoses or couplings behind the reservoirs, so the unit has to replace what the plant loses. Nothing inside the machine is faulty. The loss runs day and night, which is what separates it from a busy shift, and it grows slowly as more joints work loose. The pressure falls at the same rate at night and at the weekend, when no consumer is drawing air.	- Stop the unit, close the isolation valve at the pneumatic panel and watch whether the reservoir pressure still falls. - If it holds, open the valve again and repeat the hold test with the plant isolated section by section. - Walk the distribution network and listen at joints, couplings and hose ends, starting with the branches changed most recently. - Compare the pressure decay of a quiet night with the reference cycle of section 6.3.	Seal or replace the leaking joints, hoses and couplings, then repeat the pressure-hold test with the unit stopped and the plant isolated. Record the decay you measure so the next comparison starts from a known figure. Leaks return, so add a walk-round listening check to the routine in task 7.3.2 rather than treating this as a one-off repair.	Weekly checks; W101, S305, W108, W102
Signals: How fast line pressure falls while the unit is not delivering is gradually faster. The earliest sign, visible in the idle phase of every cycle. How often the compressor loads per hour is gradually higher. How long each loaded run lasts is gradually longer while loaded. Whether a loaded run ends at the cut-out pressure is not reached. Late in the fault, once the loss matches what the unit can deliver. Line pressure (P2) falls persistently while loaded. Late in the fault the line sinks while the unit is still delivering. Low-pressure switch (D5) is persistently on while loaded. The switch closes only at that late stage, so it gives no early warning.			
low_line_pressure	high_air_demand		distribution
Air demand above the rated delivery (common, benign) The consumers ask for more air than the unit is rated to deliver, usually after a new machine went into service or a shift pattern changed. The unit works exactly as designed and needs no repair; it simply cannot keep up with what is now being asked of it. The shortfall follows the production pattern and disappears when the plant stops.	- Ask the operators whether a consumer was added, enlarged or moved to a longer duty. - Watch the line pressure across a full shift and across a quiet night, and compare the two. - Confirm that the pressure holds when the plant is idle; a demand problem disappears then, a leak does not. - Compare the loaded time per hour with the reference duty of section 6.3.	No repair is needed on the unit. Reduce the demand, move the heaviest consumers out of the peak, or plan additional compressed-air capacity. Until then, monitor the oil temperature and the loaded time per hour: the long loaded runs are what push the temperature up, and a unit that is loaded almost continuously has no margin left for a real fault. Record the new demand so later comparisons start from it.	Weekly checks; W101, S305, W108, W102, W104, X201, S301, W109, W116, S306
Signals: How long each loaded run lasts is persistently longer while loaded. How often the compressor loads per hour is persistently higher. Line pressure (P2) is persistently low while loaded. How fast line pressure falls while the unit is not delivering is faster. The decay follows the production pattern and stops when the plant stops, unlike a leak. Oil temperature (T1) rises gradually while loaded. A consequence of the long loaded runs, not of a cooling problem.			
low_line_pressure	dryer_purge_leak		dryer
Dryer purge valve not seating (common) The purge valve no longer closes fully, so dried air escapes continuously through the regeneration line and the silencer. The unit delivers against that loss and never reaches its cut-out point, while the line stays above the low-pressure switch setting, so the controller gives no warning of a pressure problem at all. Look at the purge line before the distribution network; the loss is inside the unit.	- Listen at the purge silencer while the unit is loaded; a steady discharge between tower changeovers points at the valve. - Watch dryer purge pressure (P4) through a complete changeover and check whether it falls back to its resting value. - Isolate the dryer and see whether the unit reaches its cut-out pressure again. - Inspect the purge valve seat, spring and pilot line as described in task 7.3.10.	Fit a new purge valve kit and check the silencer for erosion while the valve is out. Refit, run a full regeneration cycle and confirm that the purge pressure returns to its resting value between changeovers and that the unit reaches its cut-out point again. A valve that has leaked for a long time usually means the silencer packing is damaged too; replace both together.	Purge valve and silencer service; W101, S305, W102, W103
Signals: Dryer purge pressure (P4) is persistently high while loaded. The clearest single sign; the purge line is pressurised continuously, not only at a changeover. Whether a loaded run ends at the cut-out pressure is not reached. How long each loaded run lasts is persistently longer while loaded. Oil temperature (T1) rises gradually while loaded. The unit is loaded for hours on end, so the oil keeps taking up heat. Motor current (I1) is persistently low while loaded. The element works against a plateau below the normal delivery pressure. Low-pressure switch (D5) stays off in every state. The line never falls far enough to close the switch, so the native warning is absent.			

Condition	Fault id		Subsystem
Possible cause	What to check	Remedy	See also
low_line_pressure	intake_filter_clogged		intake unloading
Air intake filter clogged (occasional) A blocked intake element throttles the suction side, so the compression element never receives a full charge of air. The motor turns more easily than it should, the unit delivers less per loaded run, and the restriction grows slowly over the whole service interval.	- Inspect the intake element against the light and look for an even layer of dust over the whole surface. - Read the running hours since the last intake filter service and compare them with task 7.3.3. - Check the condition of the room; a dusty site shortens the life of the element considerably.	Fit a new intake filter element, clean the housing and the intake grille, and record the service so the next interval starts from the right hour count. Confirm afterwards that the loaded current and the pressure rise per loaded run are back in their normal bands. On a dusty site, shorten the interval rather than waiting for the symptom to come back.	Air intake filter service; W101, S305, W102, W107
Signals: Motor current (I1) is gradually low while loaded. How fast line pressure rises during a loaded run is gradually slower. How long each loaded run lasts is gradually longer while loaded.			
low_line_pressure	condensate_drain_stuck_open		separator drain
Condensate drain stuck open (occasional) The automatic drain no longer closes between pulses, so compressed air escapes with the condensate. The unit replaces the loss and loads more often, and the pressure at the separator discharge collapses instead of following the line while the unit idles.	- Listen at the drain outlet between drain pulses; a steady hiss of air is the clearest sign. - Watch separator discharge pressure (P3) while the unit idles; it should match the line pressure. - Inspect the drain valve seat and the strainer for debris holding the valve off its seat. - Read the running hours since the last drain service against task 7.3.8.	Clean or rebuild the drain valve, clear its strainer and confirm that it closes tightly between pulses. Check the separator discharge pressure over a complete cycle afterwards. Where debris caused the fault, shorten the drain service interval, since the same debris will hold the rebuilt valve open just as easily.	Condensate drain service; W101, S305, W108, W112
Signals: Separator discharge pressure (P3) is persistently near zero while unloaded. It should follow the line pressure while the unit is not delivering. How often the compressor loads per hour is persistently higher. How fast line pressure falls while the unit is not delivering is faster. Dryer purge pressure (P4) is unchanged in every state. The purge line is normal, which separates this from a leaking purge valve.			
low_line_pressure	airend_element_wear		compressor
Compression element worn (occasional) The clearances inside the compression element have opened up, so part of the air slips back instead of being delivered. The unit turns easily and quietly but no longer fills the line, and the difference between the vessel and the line falls below its reference figure.	- Compare the vessel pressure with the line pressure while the unit delivers; a difference that has shrunk over months points at wear. - Rule out a clogged intake filter and a sticking intake valve first, since both also lower the loaded current. - Compare the time a loaded run takes with the reference cycle of section 6.3. - Check the running hours of the unit against the inspection interval.	Have the compression element inspected and overhauled or exchanged by a service engineer. Wear of this kind is gradual and does not repair itself, so plan the work rather than waiting for a breakdown. Change the oil and both filters after the overhaul and record the new reference figures for the loaded current and the pressure rise per loaded run.	Annual inspection; W101, S305, W102, W107
Signals: Motor current (I1) is gradually low while loaded. How fast line pressure rises during a loaded run is gradually slower. How long each loaded run lasts is gradually longer while loaded. Discharge pressure (P1) is gradually low while loaded. The difference between the vessel and the line is smaller than the reference figure.			
low_line_pressure	minimum_pressure_valve_fault		compressor
Minimum-pressure valve fault (occasional) The minimum-pressure valve no longer opens at its setting or no longer holds. Stuck shut, it lets the vessel pressure climb while the line stays low; leaking back, it lets the vessel drain into the line so that the oil circuit never gets the pressure it needs to circulate. The vessel pressure stays high while the line stays low; that split points at the valve.	- Compare the vessel pressure with the line pressure while the unit delivers; a much larger difference than usual points at this valve. - Rule out a clogged separator element first, since it produces a similar difference but builds up slowly. - Check the valve seat, spring and guide for deposits and for a sticking stem. - Look for oil starvation at the element, which a valve that leaks back will cause.	Overhaul or replace the minimum-pressure valve, renew its spring and clear the seat. Refit and confirm over a full cycle that the vessel and line pressures track each other with the normal difference between them and that the loaded current is back in its band. Have the oil circuit checked if the valve was leaking back, because the element may have run with too little oil.	Annual inspection; W101, S305, W106, X203, S303, X202, S302, W117
Signals: Discharge pressure (P1) is persistently high while loaded. With the valve stuck shut; when it leaks back the vessel pressure stays low instead. Line pressure (P2) is persistently low while loaded. Motor current (I1) is persistently high while loaded. How fast line pressure rises during a loaded run is slower. The line sees little of the air the element is actually delivering.			

Condition	Fault id		Subsystem
Possible cause	What to check	Remedy	See also
low_line_pressure	intake_valve_not_opening		intake unloading
Intake valve not opening fully (occasional) The intake and unloading valve stays partly closed when the unit loads, so the element never gets its full charge of air. The motor turns easily and the unit delivers far less than its rating. Unlike a clogged filter, the change is abrupt and the intake element is clean.	- Compare the valve feedback with the load command while the unit is delivering; a disagreement isolates the valve. - Inspect the intake element first, to rule out the far more common blockage. - Check the pilot line and the actuator of the intake valve for leaks and for a sticking stem. - Look for gum or dust deposits on the valve seat and guide.	Clean or overhaul the intake valve, free the stem and renew the pilot line if it leaks. Refit and confirm over a full cycle that the valve feedback follows the load command and that the loaded current and the pressure rise per loaded run are back in their normal bands. Fit a new intake filter element at the same time, because dust carried past a worn element is the usual reason the valve sticks.	Air intake filter service; W101, S305, W102, W107
Signals: Motor current (I1) is persistently low while loaded. The change appears from one cycle to the next rather than over weeks. How fast line pressure rises during a loaded run is persistently slower. Intake valve closed (D1) stays on while loaded. The valve feedback still reports closed while the controller is asking for load. Whether a loaded run ends at the cut-out pressure is not reached.			
low_line_pressure	unloader_solenoid_fault		intake unloading
Unloader solenoid fault (rare) The solenoid that switches the unit between load and idle no longer follows the controller. Stuck in the load position the unit compresses past its cut-out point; stuck in the idle position it runs on without ever building pressure. The coil, its wiring and the pilot air are all possible reasons. Only when the solenoid fails in the idle position, so the unit never loads.	- Compare the load command of the controller with the solenoid state while the unit cycles. - Listen at the solenoid as the controller switches; a coil that does not pick up is usually audible. - Measure the coil and check its wiring at the terminal strip. - Check the pilot air supply to the solenoid for blockage or moisture.	Replace the solenoid or repair its wiring, and clear the pilot air line of moisture and debris. Refit and confirm over several complete cycles that the unit loads at the cut-in setting and unloads at the cut-out setting. Where wet pilot air caused the fault, check the condensate drain as well, because it will otherwise ruin the new coil in the same way.	Condensate drain service; W101, S305, X204
Signals: Load solenoid (D2) stays on persistently in every state. With the solenoid stuck in the load position; stuck the other way it stays off instead. Intake valve closed (D1) stays off persistently while unloaded. Line pressure (P2) rises persistently while unloaded. Whether a loaded run ends at the cut-out pressure is not reached. With the solenoid stuck in the idle position the unit never builds pressure to the cut-out point.			
low_line_pressure	regulator_contact_fault		control
Pressure regulator contact fault (rare) The regulator contact that tells the controller whether the line pressure is satisfied no longer changes state. The controller keeps the unit in whatever mode the stuck contact reports, however the real pressure behaves, so regulation stops working while every sensor still reads correctly.	- Watch the regulator contact while the line pressure passes the cut-out setting; a contact that does not change isolates the fault. - Compare the pressure on the display with a test gauge at the panel to rule out a transducer fault. - Inspect the contact, its pilot connection and its wiring at the pneumatic panel.	Clean or replace the regulator contact and renew its pilot connection and wiring as needed. Confirm over several cycles that the contact changes state at the cut-in and cut-out settings and that the unit follows it. Until the repair is made, keep the unit under supervision: with regulation out of action the only protection left against overpressure is the safety valve.	W101, S305, X204
Signals: Pressure regulator contact (D4) stays off persistently in every state. A contact stuck in the load-requested position keeps the unit loaded; stuck the other way it keeps the unit idle. Line pressure (P2) rises persistently while loaded. Whether a loaded run ends at the cut-out pressure is not reached. The run overshoots the cut-out point instead of ending at it. Load solenoid (D2) stays on in every state.			
low_line_pressure	line_pressure_transducer_fault		control
Line pressure transducer fault (rare) The line pressure transducer or its wiring reports a value that the pipework does not have. The controller regulates on the wrong figure, so it can hold the unit loaded past the real cut-out point or unload it while the line is still empty, while every other reading stays correct. The pressure is only low on the display; a test gauge at the panel disagrees with it.	- Compare the display with a test gauge at the pneumatic panel, at rest and under load. - Compare the line reading with the reservoir reading, which shares the same air circuit. - Compare the reading with the state of the low-pressure switch, which is independent of the transducer. - Inspect the transducer, its impulse line and its wiring for moisture, blockage and loose terminals.	Clear the impulse line, repair the wiring or replace the transducer, then confirm the reading against a test gauge across the working range. Keep the unit under supervision until the reading is trusted again: with a faulty regulation signal the unit can compress past its cut-out point, and only the safety valve then stands between it and an overpressure.	Annual inspection; W101, S305, X204, W113
Signals: Line pressure (P2) fluctuates intermittently in every state. Reservoir pressure (P5) is unchanged in every state. The reservoir transducer keeps reporting the true pressure, so the two readings drift apart. Whether a loaded run ends at the cut-out pressure is not reached. Low-pressure switch (D5) stays off in every state. The mechanical switch follows the real pressure and contradicts the display.			

Condition	Fault id		Subsystem
Possible cause	What to check	Remedy	See also
8.4 Compressor starts and loads too often frequent_cycling — The unit loads and unloads much more often than the reference duty allows and message W108 (section 3.4) appears. Each loaded run is followed by a short idle phase, and the unit rarely stops. Operators notice the noise of the repeated starts before the controller reports anything.			
frequent_cycling	downstream_air_leak		distribution
Leak in the distribution network (common) The clearest early sign; the idle pressure decay is faster than the reference figure long before the pressure itself drops.	The checks of 8.3.	The remedy of 8.3.	Weekly checks; W101, S305, W108, W102
frequent_cycling	high_air_demand		distribution
Air demand above the rated delivery (common, benign)	The checks of 8.3.	The remedy of 8.3.	Weekly checks; W101, S305, W108, W102, W104, X201, S301, W109, W116, S306
frequent_cycling	condensate_drain_stuck_open		separator drain
Condensate drain stuck open (occasional)	The checks of 8.3.	The remedy of 8.3.	Condensate drain service; W101, S305, W108, W112
frequent_cycling	reservoir_isolation_valve_closed		reservoirs
Reservoir isolation valve closed (rare) An isolation valve at a reservoir has been left closed, usually after maintenance, so the unit works against a much smaller volume. It fills and empties that volume quickly, the reservoir reading no longer follows the line, and no air passes the flow sensor. Here the loaded runs get shorter rather than longer, because the unit fills a much smaller volume.	• Check the position of the isolation valve at each reservoir; this is the quickest check and the usual answer. • Compare the reservoir reading with the line reading over a complete cycle. • Ask whether any work has been done on the reservoirs or their safety valves recently.	Open the isolation valve slowly, with the unit stopped, and let the reservoir fill before starting again. Confirm that the reservoir and line readings track each other and that the flow sensor pulses while the unit delivers. Add a valve position check to the hand-over after any work on the reservoirs, because this is a procedural fault, not a technical one.	Weekly checks; W108, W113, W114
Signals: Reservoir pressure (P5) is unchanged in every state. The pressure behind the closed valve stands still while the line pressure cycles. How long each loaded run lasts is shorter while loaded. Shorter runs, unlike a leak or heavy demand, which both make runs longer. How often the compressor loads per hour is higher. Flow pulse (D8) gives no pulse persistently while loaded.			
8.5 Compressor stays loaded and does not reach cut-out continuous_load — The unit delivers without pause and message W102 (section 3.4) appears once the permitted continuous load time has passed. The line pressure settles on a plateau below the cut-out setting instead of rising to it, and the oil temperature climbs while the unit keeps running.			
continuous_load	dryer_purge_leak		dryer
Dryer purge valve not seating (common) Signature of a loss inside the unit; the purge pressure stays raised while the unit delivers and the low-pressure switch never closes.	The checks of 8.3.	The remedy of 8.3.	Purge valve and silencer service; W101, S305, W102, W103
continuous_load	downstream_air_leak		distribution
Leak in the distribution network (common) Signature of a loss in the plant; the cycles grew longer and closer for days before the unit stuck at load.	The checks of 8.3.	The remedy of 8.3.	Weekly checks; W101, S305, W108, W102
continuous_load	high_air_demand		distribution
Air demand above the rated delivery (common, benign)	The checks of 8.3.	The remedy of 8.3.	Weekly checks; W101, S305, W108, W102, W104, X201, S301, W109, W116, S306
continuous_load	intake_filter_clogged		intake unloading
Air intake filter clogged (occasional)	The checks of 8.3.	The remedy of 8.3.	Air intake filter service; W101, S305, W102, W107
continuous_load	airend_element_wear		compressor
Compression element worn (occasional)	The checks of 8.3.	The remedy of 8.3.	Annual inspection; W101, S305, W102, W107

Condition	Fault id		Subsystem
Possible cause	What to check	Remedy	See also
continuous_load	intake_valve_not_opening		intake unloading
Intake valve not opening fully (occasional)	The checks of 8.3.	The remedy of 8.3.	Air intake filter service; W101, S305, W102, W107
continuous_load	purge_silencer_damaged		dryer
Purge silencer damaged (rare) The silencer packing has broken up or blown out, so the regeneration air leaves almost unrestricted and far more loudly than usual. The purge line drains slowly after every changeover instead of clearing at once, and the unit loses more air to regeneration than it was sized for.	- Look at the silencer body for cracks, missing packing or loose material around the outlet. - Compare the noise of a regeneration pulse with the noise the unit made when it was commissioned. - Watch how long the purge pressure takes to fall back after a changeover.	Fit a new silencer and clear any packing material that has been carried into the purge line. Run a full regeneration cycle and confirm that the purge pressure falls back promptly and that the discharge noise is back to its usual level. Check the purge valve at the same time, because packing debris often damages the seat.	Purge valve and silencer service; W102, W103

Signals: Dryer purge pressure (P4) is intermittently high in every state. Raised for a long while after each changeover rather than continuously. Purge pressure switch (D6) stays on while loaded. The discharge is still being detected long after the pulse should have ended. How long each loaded run lasts is gradually longer while loaded. Whether a loaded run ends at the cut-out pressure is not reached. Only in the worst case, when the escaping regeneration air matches the delivery.

8.6 Dryer purge pressure high, air escaping at the purge silencer purge_pressure_high — message W103 (section 3.4) appears while the unit is loaded and air can be heard leaving the purge silencer between tower changeovers. The purge pressure stands well above its resting value instead of falling back after each regeneration pulse.

purge_pressure_high	dryer_purge_leak		dryer
Dryer purge valve not seating (common) The purge pressure is high continuously, not only around a changeover.	The checks of 8.3.	The remedy of 8.3.	Purge valve and silencer service; W101, S305, W102, W103
purge_pressure_high	purge_silencer_damaged		dryer
Purge silencer damaged (occasional) The discharge is much louder than usual and the purge pressure decays slowly after each pulse.	The checks of 8.5.	The remedy of 8.5.	Purge valve and silencer service; W102, W103
purge_pressure_high	tower_changeover_valve_fault		dryer
Tower changeover valve fault (occasional) A changeover valve no longer moves the drying duty from one tower to the other. One tower carries the whole load and is never regenerated, while the purge line behaves abnormally because the regeneration path is only partly open. The air leaving the dryer turns wet within hours.	- Watch the tower indication on the display over several changeover periods and note whether it alternates. - Feel both tower bodies; the tower in service is warmer than the one being regenerated, and a stuck valve leaves them alike. - Operate the changeover valves by hand where the design allows it and listen for the regeneration discharge. - Inspect the valve pilot lines and actuators for leaks or damage.	Free or replace the faulty changeover valve and its pilot line, then run several changeover periods and confirm that both towers take the duty in turn. Have the desiccant of the tower that was left in service inspected under task 7.3.9, because a charge that has run without regeneration is usually saturated and may need replacing.	Dryer desiccant inspection; W103, W111, W115

Signals: Dryer tower in service (D3) stays on persistently in every state. The tower indication stops alternating and holds one position. Dryer purge pressure (P4) is intermittently high while loaded. Purge pressure switch (D6) gives no pulse while loaded. No clean discharge is detected, because the regeneration path never opens fully. How often the compressor loads per hour is unchanged. The compressor side of the unit is untouched by this fault.

8.7 Oil temperature high oil_temperature_high — The controller shows a rising oil temperature and works up the warning chain from message W104 (section 3.4) to message X201 (section 3.4) and finally message S301 (section 3.4), which stops the unit. The machine feels hotter than usual and the cooling air leaving it is warm.

oil_temperature_high	oil_cooler_fouled		cooling
Oil cooler fouled (common) Dust and oil mist have settled on the cooler block, so the cooling air can no longer carry the heat away. The oil runs hotter in every state, while the duty of the unit and the room around it stay exactly as they were. The change comes on slowly, over weeks rather than hours. The rise is slow and shows in every state, including the idle and stopped phases.	- Look at the cooler block and the fan guard against the light for a dust or oil film between the fins. - Compare the oil temperature with the ambient temperature over a full cycle; a difference larger than usual points at the cooler. - Confirm that the fan turns freely and at its usual speed, to separate this from a fan failure. - Read the running hours since the last cooler cleaning against task 7.3.7.	Clean the cooler block, the aftercooler and the fan guard with the unit stopped and isolated, working from the clean side outwards so the dirt is not driven further in. Confirm over the next loaded runs that the oil temperature comes back to its reference figure. Where the site is dusty, shorten the cleaning interval instead of waiting for the warning to return.	Oil cooler and aftercooler cleaning; W104, X201, S301, W109, W116, S306

Signals: Oil temperature (T1) rises gradually in every state. The rise shows in the idle and stopped phases too, not only under load. Ambient temperature (T2) is unchanged in every state. The room is as warm as it always was, which rules out a hot site. How often the compressor loads per hour is unchanged. The unit is not working any harder than before. Motor current (I1) is unchanged while loaded.

Condition	Fault id		Subsystem
Possible cause	What to check	Remedy	See also
oil_temperature_high	high_ambient_temperature		cooling
Ambient temperature above the operating range (common, benign) The cooling air reaching the unit is warmer than the reference conditions allow, so the cooler cannot pull the oil down to its usual temperature. The unit and its cooling circuit are sound; the room around them has changed. Warm air also carries more moisture into the dryer. The room itself is warm; the difference between oil and cooling air is the same as always.	- Read the ambient temperature at the cooling-air inlet and compare it with the operating limits of section 5.2. - Check that the cooling-air inlet and outlet are clear and that warm outlet air is not being drawn back in. - Compare the oil temperature with the ambient temperature; when the difference between them is normal, the cooler is doing its job. - Check whether other equipment in the room has started to give off heat.	No repair is needed on the unit. Improve the ventilation of the room, duct the warm outlet air away from the intake, or move the heat source. Until the room is put right, monitor the oil temperature and the difference between it and the ambient reading: as long as that difference stays normal, the cooling circuit is sound. Watch the delivered air for moisture as well, because a warm intake loads the dryer more heavily.	Oil cooler and aftercooler cleaning; W104, X201, S301, W109, W116, S306
Signals: Ambient temperature (T2) is persistently high in every state. Oil temperature (T1) rises gradually in every state. The difference between oil and cooling air stays as it always was; both simply sit higher. How often the compressor loads per hour is unchanged. Motor current (I1) is unchanged while loaded. Nothing is dragging; this separates a warm room from mechanical wear.			
oil_temperature_high	oil_level_low		oil
Oil level below the minimum (common) Too little oil circulates, so the charge that is left has to take up the heat of the whole compression process. The temperature climbs while the unit otherwise behaves normally. A level that keeps falling always has a second cause behind it, which has to be found as well.	- Stop the unit, let the vessel vent and stand, then read the level at the sight glass. - Look under and around the machine for oil at the cooler, the hoses and the shaft seal. - Check the downstream filters for oil, which would point at carry-over rather than a leak. - Compare how much oil has been added recently with the interval in task 7.3.4.	Top the charge back up to the mark with the specified grade, then find out where it went: an external leak, carry-over through the separator, or a blocked scavenge line. Topping up alone treats the symptom only. Confirm once the level holds that the oil temperature comes back into its normal band.	Oil change, Daily checks; W104, X201, S301, W109, W116, S306
Signals: Oil level switch (D7) is persistently off in every state. Oil temperature (T1) rises gradually while loaded. How often the compressor loads per hour is unchanged.			
oil_temperature_high	oil_filter_clogged		oil
Oil filter clogged (occasional) The filter element restricts the oil flow, so less oil reaches the compression element and the injected charge carries away less heat. The restriction builds up slowly across the service interval, and nothing else on the unit changes with it.	- Read the running hours since the last oil filter change against task 7.3.5. - Feel the filter housing while the unit is loaded and compare the temperature across it. - Check the oil change record, since oil left in service too long blocks a filter early.	Fit a new oil filter element and top the charge up to the mark. Change the oil as well if it is anywhere near its interval, because a filter that blocked early is a sign of the oil, not of the filter. Confirm that the oil temperature comes back into its normal band over the next few loaded runs and record the service.	Oil filter change; W104, X201, S301, W109, W116, S306
Signals: Oil temperature (T1) rises gradually while loaded. Motor current (I1) is unchanged while loaded. How often the compressor loads per hour is unchanged.			
oil_temperature_high	cooling_fan_failure		cooling
Cooling fan failure (occasional) The cooling fan has stopped or turns far too slowly, so almost no air passes through the cooler block. The oil temperature climbs sharply as soon as the unit loads, much faster than a fouled block could explain, and it keeps climbing until the controller stops the unit. The rise sets in abruptly at the next loaded run rather than over days.	- Look and listen at the fan while the unit is loaded, and feel for air leaving the cooler outlet. - Isolate the unit and turn the fan by hand to check for a seized bearing or a fouled impeller. - Check the fan supply, its protection and its wiring at the terminal strip. - Look for anything blocking the cooling-air inlet or outlet.	Free or replace the fan and its motor, clear whatever blocked the airway and restore the supply. Confirm over the next loaded runs that the oil temperature settles at its reference figure. Clean the cooler block while the fan is out, because a unit that has run hot has usually baked oil mist onto the fins as well.	Oil cooler and aftercooler cleaning; W104, X201, S301, W109, W116, S306
Signals: Oil temperature (T1) rises suddenly while loaded. The onset is abrupt, which is what separates this from a slowly fouling cooler. Ambient temperature (T2) is unchanged in every state. How often the compressor loads per hour is unchanged. Motor current (I1) is unchanged while loaded. The compression side is unaffected; only the cooling air is missing.			
oil_temperature_high	high_air_demand		distribution
Air demand above the rated delivery (occasional, benign) The unit is simply loaded far more of the time; the cooling circuit itself is sound.	The checks of 8.3.	The remedy of 8.3.	Weekly checks; W101, S305, W108, W102, W104, X201, S301, W109, W116, S306

Condition	Fault id		Subsystem
Possible cause	What to check	Remedy	See also
oil_temperature_high	thermostatic_valve_stuck		cooling
Thermostatic valve stuck (occasional) The valve that decides how much oil passes through the cooler has stuck. Held towards the bypass, the oil never cools; held towards the cooler, it never warms up and water separates out in it. In both cases the oil temperature stops responding to the duty and to the room.	- Feel the pipework on both sides of the valve while the unit warms up; a working valve sends the oil to the cooler only once it is warm. - Watch the warm-up curve after a cold start and note whether the oil settles at its working temperature. - Rule out a fouled cooler and a failed fan first, since both are more common. - Inspect the valve element for deposits and for a stem that no longer moves.	Replace the thermostatic element and clean its housing. Confirm after a cold start that the oil warms up at its usual rate and then settles at its working temperature. Drain any condensate from the vessel if the valve was stuck towards the cooler, because oil that has run cold for a long time holds water.	Oil cooler and aftercooler cleaning; W104, X201, S301, W109, W116, S306, W105, W110
Signals: Oil temperature (T1) rises gradually while loaded. With the valve stuck towards the bypass, so the oil hardly passes through the cooler. Oil temperature (T1) is persistently low in every state. With the valve stuck towards the cooler, so the oil stays below its working temperature even after a long run. Ambient temperature (T2) is unchanged in every state. How often the compressor loads per hour is unchanged.			
oil_temperature_high	wrong_oil_grade		oil
Wrong oil grade in service (rare) The unit has been filled with an oil that is not the specified grade, or with two grades mixed together. It carries heat away less well, drags in the element and separates poorly, so temperature, current and carry-over all move at once without any single part being at fault.	- Read the service record for the grade and the quantity used at the last oil change. - Look at the oil at the sight glass for a colour or a foaming behaviour that does not match the specified grade. - Check whether the separator element blocked earlier than its interval, which the wrong grade will cause. - Compare the oil temperature and the loaded current with the figures recorded before the last oil change.	Drain the charge completely, flush the circuit, and refill with the specified grade and quantity given in section 9.1. Change the oil filter and the separator element at the same time, because the wrong grade damages both. Record the grade used so the mistake is not repeated, and confirm afterwards that the oil temperature and the loaded current return to their normal bands.	Oil change; W104, X201, S301, W109, W116, S306, W106, X203, S303
Signals: Oil temperature (T1) rises gradually while loaded. Motor current (I1) rises gradually while loaded. Oil level switch (D7) is gradually off in every state. Poor separation carries oil out with the air, so the level falls between services. How often the compressor loads per hour is unchanged.			
oil_temperature_high	oil_temperature_sensor_fault		control
Oil temperature sensor fault (rare) The temperature sensor or its wiring reports a value that the oil itself does not have. The reading jumps or drifts while every other sign of a hot or a cold machine is missing, and the controller acts on a temperature that is not there. Nothing else on the unit is hot; a hand-held thermometer contradicts the display.	- Compare the reading with a hand-held thermometer at the airend oil outlet. - Feel the cooler outlet and the vessel; a machine that is really hot cannot be mistaken by hand. - Inspect the sensor wiring and its terminals for damage, moisture and loose screws. - Watch the reading through a full cycle and look for steps that no heating or cooling could explain.	Repair the wiring or replace the sensor, then confirm the reading against a reference instrument at the working temperature. Reset the message and watch the next few cycles. Until the sensor is trusted again, do not raise the temperature settings to silence the warning: that would leave the unit without its real protection.	W104, X201, S301, W109, W116, S306, W105, W110
Signals: Oil temperature (T1) fluctuates intermittently in every state. Steps and jumps that no thermal process could produce. Ambient temperature (T2) is unchanged in every state. How often the compressor loads per hour is unchanged. Motor current (I1) is unchanged while loaded. Nothing else that would accompany a genuinely hot machine is present.			
8.8 Oil temperature stays low, condensate in the oil oil_temperature_low — message W105 (section 3.4) appears and the oil does not reach its working temperature even after a long run. The oil looks milky at the sight glass and water separates out at the drain point, because the charge never gets warm enough to drive the moisture off.			
oil_temperature_low	low_ambient_temperature		cooling
Ambient temperature below the operating range (common, benign) The unit stands in a cold room, so the oil is thick at rest and warms very slowly. Water condenses in a charge that never reaches its working temperature, and a cold start is hard enough to trip the motor protection. The machine itself is undamaged.	- Read the ambient temperature at the cooling-air inlet and compare it with the operating limits of section 5.2. - Watch how long the oil takes to reach its working temperature after a cold start. - Look at the oil through the sight glass for the milky appearance that water in the oil gives. - Check whether a door, a broken window or a failed room heater is letting cold air in.	No repair is needed on the unit. Keep the room above the lower operating limit, close the source of the cold draught, or fit room heating. Until then, monitor the oil temperature during the warm-up after each start and drain condensate from the vessel more often, because water separates out of oil that never gets warm. Report repeated hard starts, since they shorten the life of the motor.	Daily checks; W105, W110, S306, S304, S307, S308
Signals: Ambient temperature (T2) is persistently low in every state. Oil temperature (T1) is persistently low in every state. The current peak at motor start is higher at start. Thick oil makes the motor work much harder for the first turns of a start. How often the compressor loads per hour is unchanged.			
oil_temperature_low	thermostatic_valve_stuck		cooling
Thermostatic valve stuck (occasional) Here the oil stays cold whatever the room does, because the whole flow goes through the cooler.	The checks of 8.7.	The remedy of 8.7.	Oil cooler and aftercooler cleaning; W104, X201, S301, W109, W116, S306, W105, W110

Condition	Fault id		Subsystem
Possible cause	What to check	Remedy	See also
oil_temperature_low	oil_temperature_sensor_fault		control
Oil temperature sensor fault (rare)	The checks of 8.7.	The remedy of 8.7.	W104, X201, S301, W109, W116, S306, W105, W110

8.9 Motor current high motor_current_high — The current of the monitored phase runs above the normal loaded band and the controller works up from message W106 (section 3.4) through message X203 (section 3.4) to message S303 (section 3.4). The motor sounds strained and its housing is warmer than usual.

motor_current_high	separator_element_clogged		compressor
Separator element clogged (common) The separator element is loaded with dirt and oil residue, so the air has to force its way through it. The vessel works at a higher pressure than the line, the motor takes more current for the same delivery, and the element starts to let oil through with the air. The vessel pressure rises with the current while the line pressure keeps its setting.	- Compare the vessel pressure with the line pressure while the unit delivers and watch how the difference has grown over recent weeks. - Read the running hours since the last separator element change against task 7.3.6. - Check whether the oil grade and the oil change interval have been kept, since the wrong oil blocks an element quickly. - Inspect the downstream filters for oil carry-over.	Fit a new separator element and renew its seals and the scavenge line strainer while the vessel is open. Change the oil and the oil filter at the same time if the element blocked early, because that is almost always a sign of the wrong grade or an overdue oil change. Confirm afterwards that the difference between the vessel and line pressures is back to its reference figure.	Separator element change; W106, X203, S303, X202, S302, W117, W116

Signals: Discharge pressure (P1) is gradually high while loaded. Motor current (I1) is gradually high while loaded. Line pressure (P2) is unchanged while loaded. The line still follows its setting; it is the difference across the element that grows. Oil temperature (T1) rises gradually while loaded.

motor_current_high	intake_valve_not_closing		intake unloading
Intake valve not closing (occasional) The intake valve does not shut when the controller unloads, so the unit keeps drawing and compressing air past its cut-out point. Pressure and current stay at their loaded values through what should be the idle phase, and the safety valve can lift. The current stays at load level during the idle phase as well, which no mechanical drag explains.	- Stop the unit at once if the safety valve is lifting, then investigate. - Watch the valve feedback at the moment the controller unloads; a feedback that does not change isolates the valve. - Check the pilot line, the actuator and the return spring of the intake valve. - Inspect the valve seat for deposits or damage that would hold it off its seat.	Overhaul or replace the intake valve, renew the return spring and the pilot line, and clear any deposits from the seat. Refit and confirm over several cycles that the unit unloads at its cut-out setting and that the vessel vents afterwards. Have the safety valve tested under task 7.3.11 if it lifted, because a valve that has blown does not always reseal cleanly.	Safety valve test; W106, X203, S303, X202, S302, W117, X204

Signals: Intake valve closed (D1) stays off persistently while unloaded. The feedback shows the valve open when the unit should be idling. Discharge pressure (P1) is persistently high while unloaded. Motor current (I1) is persistently high while unloaded. The motor keeps doing compression work during a phase in which it should be running light. Line pressure (P2) rises persistently while unloaded. The line climbs past the cut-out setting instead of settling.

motor_current_high	supply_voltage_low_or_unbalanced		electrical
Supply voltage low or unbalanced (occasional) The supply is weak, or the phases are out of balance, so the motor draws more current for the same work and heats up. A bad start can make the protection drop out before the unit reaches speed. The extra heat is in the motor, not in the oil, which is what gives it away.	- Measure the supply voltage at the terminals while the unit is loaded, not only at rest. - Compare the three phases against each other under load and look for an imbalance. - Feel the motor housing and the terminal box; a supply problem heats the motor while the oil stays normal. - Inspect the supply cable, the fuses and the terminal connections for damage and heat marks.	Have the supply, the protection and the connections checked and put right by an electrician; nothing on the compressor side needs work. Confirm after the repair that the loaded current and the start peak are back in their normal bands. A motor that has run for a long time on an unbalanced supply should be checked before the unit returns to full duty.	Annual inspection; W106, X203, S303, S304, S307, S308

Signals: Motor current (I1) rises persistently while loaded. The current peak at motor start is higher at start. Oil temperature (T1) is unchanged in every state. The oil stays in its normal band, unlike every mechanical reason for a high current. How often the compressor loads per hour is unchanged.

motor_current_high	blowdown_valve_fault		intake unloading
Blow-down valve fault (occasional) The blow-down valve does not vent the separator vessel when the unit idles or stops. The element stays under pressure, so the next start is made against a full vessel and the motor draws far more current than it should. A vessel that is never relieved also keeps its heat. The peak appears at the start, against a vessel that was never vented.	- Watch discharge pressure (P1) at the moment the unit unloads and again when it stops; it should fall away in both cases. - Listen at the blow-down outlet for the vent that should follow every unload. - Inspect the valve, its pilot line and its silencer for blockage or a sticking stem. - Check the condensate drain, because carried-over water is what usually blocks the vent path.	Free or replace the blow-down valve and clear its pilot line and outlet. Confirm that the vessel vents within the usual time after every unload and after a stop, and that the start current is back in its normal band. Where the motor protection has dropped out repeatedly, have the motor and its protection checked before the unit goes back into normal service.	Condensate drain service; W106, X203, S303, S304, S307, S308, X204

Signals: Discharge pressure (P1) does not vent persistently while unloaded. The vessel should fall towards zero as soon as the unit stops delivering. The current peak at motor start is higher at start. Motor current (I1) is suddenly high at start. The peak is high enough to drop the motor protection out before the unit reaches speed. Oil temperature (T1) rises gradually while unloaded. The vessel stays hot because it is never relieved.

Condition	Fault id		Subsystem
Possible cause	What to check	Remedy	See also
motor_current_high	minimum_pressure_valve_fault		compressor
Minimum-pressure valve fault (occasional)	The checks of 8.3.	The remedy of 8.3.	Annual inspection; W101, S305, W106, X203, S303, X202, S302, W117
motor_current_high	airend_bearing_wear		compressor
Airend bearing wear (rare) Worn bearings in the compression element add friction to every turn. The motor has to work harder, the oil takes up the extra heat, and the running noise changes. In the end the element can become too stiff for the motor to start at all.	- Listen at the compression element for a changed running noise, and compare it with how the unit sounded when it was commissioned. - Compare the loaded current with the reference figure of section 9.1. - Rule out a clogged separator element and the wrong oil grade first, since both raise the current as well. - Check the oil for metal particles at the filter change.	Stop the unit and have the compression element inspected and overhauled by a service engineer; a bearing that is allowed to run to failure destroys the rotors as well. Check the drive coupling at the same time under task 7.3.12. Change the oil and both filters after the overhaul, because metal particles will have reached the whole circuit.	Drive coupling inspection, Annual inspection; W106, X203, S303, S304, S307, S308
Signals: Motor current (I1) rises gradually while loaded. Oil temperature (T1) rises gradually while loaded. The current peak at motor start is gradually higher at start. How fast line pressure rises during a loaded run is slower. Worn clearances let compressed air slip back through the element.			
motor_current_high	wrong_oil_grade		oil
Wrong oil grade in service (rare)	The checks of 8.7.	The remedy of 8.7.	Oil change; W104, X201, S301, W109, W116, S306, W106, X203, S303
8.10 Motor current low under load, delivery low motor_current_low — message W107 (section 3.4) appears while the unit is loaded: the motor draws less current than the normal loaded band, because the element is not compressing a full charge of air. The line pressure rises slowly and the unit runs quietly.			
motor_current_low	intake_filter_clogged		intake unloading
Air intake filter clogged (common) The element is visibly dirty and the change came on over the whole service interval.	The checks of 8.3.	The remedy of 8.3.	Air intake filter service; W101, S305, W102, W107
motor_current_low	intake_valve_not_opening		intake unloading
Intake valve not opening fully (occasional) The change was abrupt and the valve feedback contradicts the load command.	The checks of 8.3.	The remedy of 8.3.	Air intake filter service; W101, S305, W102, W107
motor_current_low	airend_element_wear		compressor
Compression element worn (rare)	The checks of 8.3.	The remedy of 8.3.	Annual inspection; W101, S305, W102, W107
8.11 Discharge pressure high, safety valve blows discharge_pressure_high — The pressure in the separator vessel runs above the line pressure by more than the normal element difference; message W117 (section 3.4) and then message X202 (section 3.4) appear, and message S302 (section 3.4) stops the unit. The safety valve may lift and vent the vessel audibly.			
discharge_pressure_high	separator_element_clogged		compressor
Separator element clogged (common)	The checks of 8.9.	The remedy of 8.9.	Separator element change; W106, X203, S303, X202, S302, W117, W116
discharge_pressure_high	minimum_pressure_valve_fault		compressor
Minimum-pressure valve fault (occasional) The line pressure stays low while the vessel pressure climbs, which a clogged element does not do.	The checks of 8.3.	The remedy of 8.3.	Annual inspection; W101, S305, W106, X203, S303, X202, S302, W117
discharge_pressure_high	intake_valve_not_closing		intake unloading
Intake valve not closing (occasional) The vessel pressure keeps climbing after the controller has asked the unit to unload.	The checks of 8.9.	The remedy of 8.9.	Safety valve test; W106, X203, S303, X202, S302, W117, X204

Condition	Fault id		Subsystem
Possible cause	What to check	Remedy	See also
8.12 Separator discharge pressure abnormal separator_pressure_abnormal – The pressure at the cyclonic separator discharge no longer follows the pattern of a normal cycle, and message W112 (section 3.4) appears while the unit is loaded. Instead of falling away under load and matching the line when the unit idles, it stays at the wrong level throughout.			
separator_pressure_abnormal	condensate_drain_stuck_open		separator drain
Condensate drain stuck open (common) The reading collapses towards zero while the unit idles, and air is heard at the drain outlet.	The checks of 8.3.	The remedy of 8.3.	Condensate drain service; W101, S305, W108, W112
separator_pressure_abnormal	condensate_drain_blocked		separator drain
Condensate drain blocked (occasional) The drain no longer opens, so condensate stands in the separator and is carried forward into the dryer. The towers take far more water than they were sized for, the desiccant is overwhelmed, and moisture reaches the consumers while every pressure still looks plausible. The reading stays high under load and no condensate leaves the drain at all.	• Watch the drain through several drain intervals and confirm whether any condensate leaves it at all. • Open the manual drain and see how much water has collected. • Inspect the drain valve, its strainer and its outlet line for sludge, scale and ice. • Check the delivered air at the consumers for water, which is the consequence to expect.	Clear or rebuild the drain valve and its outlet line, then confirm over several intervals that condensate leaves the unit normally. Have the dryer desiccant inspected under task 7.3.9, because a charge that has been flooded rarely recovers and will keep passing wet air until it is replaced.	Condensate drain service, Dryer desiccant inspection; W112
Signals: Separator discharge pressure (P3) is persistently high while loaded. It should fall away while the unit delivers, and standing condensate keeps it up. Dryer tower in service (D3) toggles in every state. The dryer keeps changing over correctly; it simply cannot cope with the water reaching it. How often the compressor loads per hour is unchanged.			
8.13 Condensate or moisture in the delivered air water_in_air – Water collects in the filters, hoses and tools of the consumers, and the air leaving the unit feels damp. The controller often reports nothing at all, because every pressure and temperature is inside its normal band; the dew point is not measured.			
water_in_air	desiccant_exhausted		dryer
Desiccant charge exhausted (common) The desiccant has taken up more water than regeneration removes, or it has simply aged past its service life. The towers still change over correctly and every pressure on the display looks normal, yet the dew point of the delivered air climbs and water reaches the consumers. The towers change over correctly and nothing else moves; only the delivered air is wet.	• Read the running hours of the desiccant charge against the interval in task 7.3.9. • Confirm that the towers change over and that a regeneration discharge follows each changeover. • Check the colour and the dust content of the charge through the inspection port. • Check that the condensate drain upstream of the dryer is working, because a flooded dryer exhausts a sound charge quickly.	Replace the desiccant charge in both towers and clean the tower screens while they are open. Put the cause of the overload right at the same time: a blocked separator drain, a warm intake or a missed regeneration will exhaust the new charge just as fast. Record the change so the next inspection interval starts from the correct date.	Dryer desiccant inspection
Signals: Dryer tower in service (D3) toggles in every state. The changeover sequence itself is still exactly as it should be. Dryer purge pressure (P4) is unchanged in every state. Regeneration still runs; it is the charge that can no longer hold the water. How often the compressor loads per hour is unchanged. Ambient temperature (T2) is persistently high in every state. A warm intake carries more moisture into the towers and shortens the life of the charge.			
water_in_air	condensate_drain_blocked		separator drain
Condensate drain blocked (common) The dryer receives far more water than it was sized for, because the separator no longer drains.	The checks of 8.12.	The remedy of 8.12.	Condensate drain service, Dryer desiccant inspection; W112
water_in_air	tower_changeover_valve_fault		dryer
Tower changeover valve fault (occasional)	The checks of 8.6.	The remedy of 8.6.	Dryer desiccant inspection; W103, W111, W115
water_in_air	high_ambient_temperature		cooling
Ambient temperature above the operating range (occasional, benign) Warm, humid intake air carries more water into the towers than the regeneration cycle removes.	The checks of 8.7.	The remedy of 8.7.	Oil cooler and aftercooler cleaning; W104, X201, S301, W109, W116, S306

Condition	Fault id		Subsystem
Possible cause	What to check	Remedy	See also
water_in_air	dryer_controller_fault		dryer
Dryer control fault (rare) The dryer control no longer issues the changeover command, so the towers keep their duty far past the intended period. Regeneration never starts, the purge line stays quiet and moisture passes through to the consumers, although the valves themselves are sound.	- Check that the dryer control has power and that its changeover period is still set as commissioned. - Watch the purge line through more than one changeover period and confirm that no discharge happens at all. - Operate a changeover valve by hand; if the towers then swap cleanly, the valves are sound and the command is missing. - Inspect the wiring between the control and the changeover valves for damage at the terminals.	Repair the wiring or replace the dryer control, then confirm over several periods that the towers alternate and that a regeneration discharge follows every changeover. Have the desiccant inspected afterwards, because a charge that has been left without regeneration holds far more water than the following cycles can drive out.	Dryer desiccant inspection; W111 , W115
Signals: Dryer tower in service (D3) stays on persistently in every state. Dryer purge pressure (P4) is persistently near zero while loaded. No regeneration flow at all, which separates this from a stuck valve. Purge pressure switch (D6) gives no pulse persistently while loaded. How often the compressor loads per hour is unchanged.			
8.14 Oil carry-over, high oil consumption oil_in_air — Oil appears in the downstream filters and at the consumers, and the unit needs topping up between oil changes. message W116 (section 3.4) eventually appears as the level falls below the minimum, although no leak is visible under the machine.			
oil_in_air	separator_element_clogged		compressor
Separator element clogged (common) Carry-over here comes with a growing difference between the vessel and the line pressure.	The checks of 8.9.	The remedy of 8.9.	Separator element change; W106 , X203 , S303 , X202 , S302 , W117 , W116
oil_in_air	oil_level_high		oil
Oil level above the mark (common) The vessel has been filled above the mark, so oil reaches the separator element faster than it can be separated out. Oil leaves with the air even though every part of the unit is sound. This is the first thing to rule out after any oil change or top-up. Always the first thing to check after an oil change or a top-up.	- Stop the unit, let the vessel vent and stand, then read the level at the sight glass. - Compare the quantity recorded at the last oil change with the fill volume in section 9.1. - Ask whether anyone has topped the unit up since the last service.	Drain the excess oil down to the mark with the unit stopped and the vessel vented, then run the unit and check the level again once it has settled. Confirm that the carry-over stops. Record the quantity that was drained, and note in the service record how much oil the unit actually takes, so the next top-up does not repeat the mistake.	Oil change, Daily checks; W116
Signals: Oil level switch (D7) stays on in every state. The switch only reports the minimum, so an overfill never shows on it. Discharge pressure (P1) is unchanged while loaded. Oil temperature (T1) is unchanged in every state.			
oil_in_air	separator_element_damaged		compressor
Separator element damaged (occasional) The element is split, or its seal has failed, so part of the air bypasses the separating media entirely. Oil leaves with the air without the difference across the element growing at all, and the level in the vessel falls much faster than the service schedule expects. Heavy carry-over with no rise in the vessel pressure at all.	- Compare the vessel pressure with the line pressure; heavy carry-over with a normal difference between them points at a damaged element rather than a clogged one. - Check how much oil has been added since the last oil change and compare it with the fill volume. - Inspect the downstream filters and the pipework for oil. - Open the vessel and examine the element and its seal for splits and for tracking marks.	Fit a new separator element with new seals and top the oil back up to the mark with the specified grade. Confirm that the carry-over stops and that the oil level then holds between services. Check how the element was damaged before closing the vessel: a collapsed element usually means it was already blocked, and refitting into the same conditions will destroy the new one as well.	Separator element change; W116
Signals: Discharge pressure (P1) is unchanged while loaded. A torn element does not restrict the flow, so the vessel pressure looks entirely normal. Oil level switch (D7) is gradually off in every state. The level switch eventually reports that the charge has fallen below the minimum. Motor current (I1) is unchanged while loaded.			
oil_in_air	scavenge_line_blocked		oil
Scavenge line blocked (occasional) The line that returns collected oil from the bottom of the separator element to the compression element is blocked. Oil gathers in the element until it is carried out with the air, although the element itself is undamaged and the pressures on the display look normal.	- Rule out an overfilled vessel and a damaged element first; both are more common and easier to see. - Check the scavenge line, its strainer and its orifice for sludge and carbon deposits. - Feel the scavenge line while the unit is loaded; a line that is returning oil is warm along its length. - Check the oil change record, since sludge comes from oil left in service too long.	Clear or replace the scavenge line, its strainer and its orifice, then change the oil and the oil filter so the new line does not block again on the same sludge. Confirm afterwards that the carry-over stops and that the oil level holds. Shorten the oil change interval if the deposits were heavy.	Oil change; W116
Signals: Oil level switch (D7) is gradually off in every state. Discharge pressure (P1) is unchanged while loaded. Nothing restricts the air path, so the vessel pressure is exactly where it should be. Oil temperature (T1) rises gradually while loaded. Less oil circulates through the element, so it carries away less heat.			

Condition	Fault id		Subsystem
Possible cause	What to check	Remedy	See also
oil_in_air	wrong_oil_grade		oil
Wrong oil grade in service (rare)	The checks of 8.7.	The remedy of 8.7.	Oil change; W104, X201, S301, W109, W116, S306, W106, X203, S303
8.15 Compressor does not start no_start — The motor does not turn when the unit is started. The controller either shows message S307 (section 3.4) or message S308 (section 3.4) at once, or it energises the load solenoid, sees no current and stops with message S304 (section 3.4). Start stays inhibited until the message is reset.			
no_start	emergency_stop_active		control
Emergency stop active (common) The emergency-stop button is still pressed, or it was never released after an earlier stop. The controller keeps the start inhibited and shows the corresponding message until the button is freed and the message is reset. It is the cheapest possible check and the most commonly missed one. Always check this first; it costs nothing and explains most sudden no-start calls.	- Look at every emergency-stop button on the unit and on any remote station and check that each is released. - Read the message on the display, which names this state directly. - Check the wiring of the emergency-stop circuit if every button is released and the message stays.	Establish why the unit was stopped before releasing anything. Once it is safe, turn the button to release it, reset the message and start the unit under supervision. Where no button is pressed, the fault is in the emergency-stop circuit itself and must be repaired by a qualified person; that circuit is a safety function and must never be bridged.	S304, S307, S308
Signals: Motor current (I1) is persistently near zero while the unit is off. Load solenoid (D2) is persistently off while the unit is off. Intake valve closed (D1) stays on persistently while the unit is off.			
no_start	motor_overload_relay_tripped		electrical
Motor protection relay tripped (common) The motor protection has dropped out and holds the start circuit open. The controller cannot start the unit until the relay is reset, and resetting it without finding the reason for the trip simply repeats the event, usually within a few cycles.	- Look at the protection relay in the cabinet and confirm that it has tripped. - Read the recorded current of the last loaded runs before the trip and look for a rise. - Check the supply voltage and the balance of the phases before resetting anything. - Turn the compression element by hand, with the unit isolated, to rule out a seizure.	Find and remove the reason for the trip first, then reset the relay and start the unit under supervision. A high current from a clogged separator element, a blow-down valve that never vents, a weak supply or worn bearings all trip the same relay, and each has its own repair. Repeated trips with no cause found mean the motor and the relay setting themselves must be checked.	Annual inspection; S304, S307, S308
Signals: Motor current (I1) is persistently near zero while the unit is off. Load solenoid (D2) is persistently off while the unit is off. The current peak at motor start is not reached at start. No start current appears at all, because the circuit never closes.			
no_start	supply_voltage_low_or_unbalanced		electrical
Supply voltage low or unbalanced (occasional)	The checks of 8.9.	The remedy of 8.9.	Annual inspection; W106, X203, S303, S304, S307, S308
no_start	blowdown_valve_fault		intake unloading
Blow-down valve fault (occasional) The unit is being asked to start against a vessel that still holds pressure.	The checks of 8.9.	The remedy of 8.9.	Condensate drain service; W106, X203, S303, S304, S307, S308, X204
no_start	low_ambient_temperature		cooling
Ambient temperature below the operating range (occasional, benign) Cold, thick oil makes the first turns so heavy that the protection drops out.	The checks of 8.8.	The remedy of 8.8.	Daily checks; W105, W110, S306, S304, S307, S308
no_start	airend_bearing_wear		compressor
Airend bearing wear (rare)	The checks of 8.9.	The remedy of 8.9.	Drive coupling inspection, Annual inspection; W106, X203, S303, S304, S307, S308
8.16 Compressor does not unload at cut-out no_unload — The line pressure passes the cut-out setting and keeps rising, and message X204 (section 3.4) appears. The unit goes on compressing, the vessel is not vented and the safety valve can lift. Pressing stop is the safe immediate action.			
no_unload	unloader_solenoid_fault		intake unloading
Unloader solenoid fault (common)	The checks of 8.3.	The remedy of 8.3.	Condensate drain service; W101, S305, X204

Condition	Fault id		Subsystem
Possible cause	What to check	Remedy	See also
no_unload	intake_valve_not_closing		intake unloading
Intake valve not closing (common) The solenoid drops out correctly but the valve itself stays open, so air keeps coming in.	The checks of 8.9.	The remedy of 8.9.	Safety valve test; W106, X203, S303, X202, S302, W117, X204
no_unload	regulator_contact_fault		control
Pressure regulator contact fault (occasional)	The checks of 8.3.	The remedy of 8.3.	W101, S305, X204
no_unload	blowdown_valve_fault		intake unloading
Blow-down valve fault (occasional) Here the unit does stop taking in air, but the vessel is never relieved.	The checks of 8.9.	The remedy of 8.9.	Condensate drain service; W106, X203, S303, S304, S307, S308, X204
no_unload	line_pressure_transducer_fault		control
Line pressure transducer fault (rare) The controller never sees the cut-out point, because the reading lags the real pressure.	The checks of 8.3.	The remedy of 8.3.	Annual inspection; W101, S305, X204, W113

8.17 Reservoir pressure differs from line pressure reservoir_deviation — message W113 (section 3.4) appears: the reservoir reading and the line reading no longer agree, although both are taken from the same air circuit. The unit otherwise runs normally and the consumers may notice nothing.

reservoir_deviation	reservoir_isolation_valve_closed		reservoirs
Reservoir isolation valve closed (common) The reservoir reading stands still while the line reading cycles.	The checks of 8.4.	The remedy of 8.4.	Weekly checks; W108, W113, W114
reservoir_deviation	reservoir_pressure_transducer_fault		control
Reservoir pressure transducer fault (occasional) The reservoir transducer reports a pressure the vessels do not have, while the line reading stays correct. A deviation appears on the display although nothing in the air circuit has changed, and the regulation of the unit is not affected at all. A test gauge at the reservoir agrees with the line, not with the display.	- Compare the display with a test gauge at the reservoir. - Check the position of the reservoir isolation valves, which produce a real deviation rather than a false one. - Inspect the transducer, its impulse line and its wiring for moisture, blockage and loose terminals.	Clear the impulse line, repair the wiring or replace the transducer, then confirm the reading against a test gauge and against the line reading with the isolation valves open. The unit can stay in service in the meantime, because regulation does not depend on this reading, but the deviation message will keep appearing until the repair is made.	Annual inspection; W113

Signals: Reservoir pressure (P5) fluctuates intermittently in every state. Line pressure (P2) is unchanged in every state. Regulation is untouched, because it does not use this reading. How often the compressor loads per hour is unchanged. Flow pulse (D8) toggles while loaded. Air is still reaching the reservoirs, so the flow sensor keeps pulsing normally.

reservoir_deviation	line_pressure_transducer_fault		control
Line pressure transducer fault (occasional) Here the regulation itself also misbehaves, because the controller regulates on the faulty reading.	The checks of 8.3.	The remedy of 8.3.	Annual inspection; W101, S305, X204, W113

8.18 Dryer towers do not change over dryer_changeover_fault — message W111 (section 3.4) or message W115 (section 3.4) appears while the unit is loaded: the controller no longer sees the tower change or the discharge that goes with it. The tower indication on the display stops alternating and the air leaving the dryer becomes wet in the hours that follow.

dryer_changeover_fault	tower_changeover_valve_fault		dryer
Tower changeover valve fault (common) The purge line still shows pressure, but the duty never moves to the other tower.	The checks of 8.6.	The remedy of 8.6.	Dryer desiccant inspection; W103, W111, W115
dryer_changeover_fault	dryer_controller_fault		dryer
Dryer control fault (occasional) No purge flow at all, because the changeover command is never issued.	The checks of 8.13.	The remedy of 8.13.	Dryer desiccant inspection; W111, W115

Condition	Fault id		Subsystem
Possible cause	What to check	Remedy	See also
dryer_changeover_fault	purge_switch_fault		dryer
Purge pressure switch fault (occasional) The switch that confirms the tower discharge no longer reports, although the towers change over and the purge line works as it should. The controller cannot see the regeneration pulse and reports a changeover problem that the dryer does not have. The dryer works correctly; only the confirmation of the discharge is missing.	• Watch dryer purge pressure (P4) and the purge switch together through one changeover; a pulse on the pressure with no switch signal isolates the switch. • Listen at the silencer to confirm that the discharge really happens. • Inspect the switch wiring and its connection at the controller terminals.	Repair the wiring or replace the pressure switch, then confirm over a complete changeover that the switch reports at the same moment as the purge pressure pulse. Until the repair is made the dryer is still drying, so the air quality is not at risk; only the supervision of it is.	W111, W115
Signals: Purge pressure switch (D6) gives no pulse persistently while loaded. Dryer purge pressure (P4) is unchanged in every state. The purge line still pulses at every changeover, which is what rules the dryer itself out. Dryer tower in service (D3) toggles in every state.			
8.19 Flow signal missing no_flow_signal — message W114 (section 3.4) appears while the unit is loaded, because no flow pulse reaches the controller from the reservoir inlet. The pressures on the display can look completely normal, so the message is easy to dismiss.			
no_flow_signal	flow_sensor_fault		reservoirs
Flow sensor fault (common) The flow sensor at the reservoir inlet, or its wiring, no longer delivers pulses although air is passing. The controller reports a missing flow signal while every pressure on the unit behaves entirely normally, so the message is easy to dismiss as a nuisance. The reservoirs keep filling, which proves that air is passing the sensor.	• Confirm that the reservoir pressure follows the line pressure, which proves that air is passing the sensor. • Check the position of the reservoir isolation valves, since a closed valve gives the same missing signal for a real reason. • Inspect the sensor, its wiring and its terminals at the controller. • Check the sensor body for debris or water that would stop it from responding.	Clean or replace the flow sensor and repair its wiring, then confirm while the unit delivers that the pulses return. The unit can stay in service in the meantime, because nothing in the air circuit is at fault, but the supervision of the flow is lost until the repair is made and the message will keep appearing.	Annual inspection; W114
Signals: Flow pulse (D8) gives no pulse persistently while loaded. Reservoir pressure (P5) is unchanged in every state. The reservoirs still fill, which proves that air is reaching them. Line pressure (P2) is unchanged in every state. How often the compressor loads per hour is unchanged.			
no_flow_signal	reservoir_isolation_valve_closed		reservoirs
Reservoir isolation valve closed (occasional) Here no air passes at all, and the reservoir pressure confirms it.	The checks of 8.4 .	The remedy of 8.4 .	Weekly checks; W108, W113, W114

8.20 After a repair

Clear the cause first and the message second. A warning clears itself once the reading has come back inside its band, a shutdown has to be acknowledged with the reset key after the cause has been removed, and a service message is cleared by resetting its counter; [3.5⁶¹](#) gives the key sequence for each of them. A message that returns within the next cycle means the cause is still there, whatever was changed, and the right response is to go back to the table rather than to reset it again.

Then run the unit through one complete cycle before you leave it: start, load, deliver until it reaches the cut-out setting, unload, let the run-on time expire and let the unit stop by itself. Watch the dryer change tower during that cycle and confirm that regeneration is audible at the silencer and stops cleanly at the end of the purge phase.

Compare every reading you noted at the start with the figures in [9.3⁶²](#), in the state each band belongs to. A repair is finished when the readings sit inside their bands, the loaded run time and the number of load cycles match the reference duty of [6.3⁶³](#), and no message stands on the display.

Close the operating log last. Record the condition as reported, the checks carried out and what they showed, the cause confirmed, the parts fitted and the readings before and after. The next person to meet this unit will read that entry before they read this chapter.

61. Section 3.5, Acknowledging and resetting messages

62. Section 9.3, Normal bands

63. Section 6.3, The normal load cycle

9 Technical data and signal list

9.1 Technical data

The ratings below apply at the reference conditions given in 5.2⁶⁴. Free air delivery, power consumption and pressure dew point are all measured there, and all three fall away as the inlet air becomes warmer, thinner or wetter, so compare a measurement on site with the reference conditions before you judge it against the table.

Technical data

Item	Value
Ratings	
Motor power	4 kW (5.36 hp)
Motor rated current	7 A
Motor speed	2950 rpm
Supply voltage	400 V
Supply frequency	50 Hz
Supply fuse	16 A
Free air delivery	0.45 m ³ /min (15.9 cfm)
Max working pressure	11 bar (160 psi)
Safety valve setting	12 bar (174 psi)
Minimum pressure valve opening	see text
Oil fill volume	8 L (2.11 US gal)
Reservoir volume each	200 L (52.8 US gal)
Reservoir count	2
Reservoir design pressure	12 bar (174 psi)
Dryer dew point	-40 °C (-40 °F)
Dryer purge fraction	see text
Desiccant charge each	3.5 kg (7.72 lb)
Sound pressure level	68 dB(A)
Length	1200 mm (47.2 in)
Width	800 mm (31.5 in)
Height	1350 mm (53.1 in)
Mass	320 kg (705 lb)
Limits	
Ambient operating	2.0–40.0 °C (36–104 °F)
Ambient storage	-20.0–60.0 °C (-4–140 °F)
Max altitude	1000 m (3281 ft)
Max relative humidity	90 %
Supply voltage tolerance	10 %
Reference conditions	
Ambient temperature T ₂	20 °C (68 °F)
Inlet pressure absolute	1 bar (14.5 psi)
Relative humidity	0 %
Cooling air temperature	20 °C (68 °F)

9.2 Signal list

Every reading the CTRL-7 shows carries a short panel label: P for a pressure, T for a temperature, I for a motor current and D for a switched input. The same label appears on the schematic, on the display, in the fault tables of chapter 8⁶⁵ and in the log the controller writes, so one label identifies a reading everywhere in this manual.

⁶⁴. Section 5.2, Reference conditions

⁶⁵. Chapter 8, Problem solving

The controller samples every signal at a fixed interval and stores it together with the machine state that was active at the time. That is why the bands in the next section are given per state and why a recorded fault can still be read back afterwards: the record keeps the reading and the context in which it was taken.

Signal list

Tag	MetroPT column	Description	Unit	Range	Sample rate	Normal loaded	Normal unloaded	Normal off	Register
P1 discharge_pressure	TP2	Discharge pressure — Pressure in the oil separator vessel, immediately downstream of the air end discharge. While the unit delivers air it follows the line pressure with the pressure drop of the separator element added on top. The blow-down valve vents the vessel as soon as the unit unloads or stops, so outside a loaded run the reading returns to atmospheric pressure.	bar	-1.0–16.0 bar (-14.5–232 psi)	10 s	8.8–10.4 bar (128–151 psi)	-0.1–0.0 bar (-1.45–0.00 psi)	-0.1–0.0 bar (-1.45–0.00 psi)	0 int16 ×1000
P2 line_pressure	TP3	Line pressure — Pressure of the compressed air in the distribution line, measured at the pneumatic panel. This is the regulation signal: the controller loads the compressor when the line falls to the cut-in setting and unloads it again when the line reaches the cut-out setting. Because the reservoirs damp short demand peaks, it is the steadiest pressure on the unit.	bar	-1.0–16.0 bar (-14.5–232 psi)	10 s	8.6–10.1 bar (125–146 psi)	9.0–10.0 bar (131–145 psi)	8.1–9.4 bar (117–136 psi)	1 int16 ×1000
P3 separator_discharge_pressure	H1	Separator discharge pressure — Pressure at the discharge port of the cyclonic separator, between the aftercooler and the dryer. While the unit is unloaded or stopped the drain is closed and the port stands at line pressure, so the reading tracks the line. While the unit delivers air the drain opens, the port is vented and the reading falls close to atmospheric pressure.	bar	-1.0–16.0 bar (-14.5–232 psi)	10 s	-0.1–0.0 bar (-1.45–0.00 psi)	9.0–10.0 bar (131–145 psi)	8.1–9.4 bar (117–136 psi)	2 int16 ×1000
P4 dryer_purge_pressure	DV_pressure	Dryer purge pressure — Pressure in the regeneration line of the desiccant dryer, downstream of the purge valve. In normal operation the line is open to the purge silencer, so the reading rests close to atmospheric pressure. Every tower changeover depressurises the off-line tower and produces a short spike.	bar	-1.0–16.0 bar (-14.5–232 psi)	10 s	-0.1–0.0 bar (-1.45–0.00 psi)	-0.1–0.0 bar (-1.45–0.00 psi)	-0.1–0.0 bar (-1.45–0.00 psi)	3 int16 ×1000
P5 reservoir_pressure	Reservoir	Reservoir pressure — Pressure in the air reservoirs, measured downstream of the second vessel. With the reservoir isolation valve open it is normally equal to the line pressure. A lasting difference between the two readings points to a restriction between the reservoirs and the pneumatic panel.	bar	-1.0–16.0 bar (-14.5–232 psi)	10 s	8.6–10.1 bar (125–146 psi)	9.0–10.0 bar (131–145 psi)	8.1–9.4 bar (117–136 psi)	4 int16 ×1000
T1 oil_temperature	Oil_temperature	Oil temperature — Temperature of the compressor oil at the air end outlet, upstream of the thermostatic valve. The thermal mass of the oil charge makes it move far more slowly than the pressures, so the bands of the three machine states overlap. A sustained rise above the normal band is the first sign of poor cooling or of a restricted oil circuit.	°C	-20.0–120.0 °C (-4–248 °F)	10 s	48.0–61.0 °C (118–142 °F)	51.0–64.0 °C (124–147 °F)	50.0–61.0 °C (122–142 °F)	5 int16 ×100

Tag	MetroPT column	Description	Unit	Range	Sample rate	Normal loaded	Normal unloaded	Normal off	Register
I1 motor_current	Motor_current	Motor current — Current of one motor phase, taken from a current transformer in the starter. It is highest while the unit is loaded, settles at a lower value while the motor runs unloaded, and falls to the residual reading of the transformer once the motor has stopped. The controller uses it to decide whether the motor is running.	A	0.0–20.0 A	10 s	5.8–6.3 A	3.6–4.0 A	0.0–0.1 A	6 int16 ×100
D1 intake_closed	COMP	Intake valve closed — Position feedback of the intake and unloading valve. The active state means the valve is closed and the airend draws in no air, which is the case while the unit is stopped and while it runs unloaded. It clears as soon as the controller loads the compressor.		—	10 s	0	1	1	7 uint16 ×1
D2 load_valve	DV_electrical	Load solenoid — Command that the controller sends to the load solenoid of the intake and unloading valve. While the solenoid is energised the intake valve is open and the unit compresses. Together with the intake valve feedback it defines the loaded machine state.		—	10 s	1	0	0	8 uint16 ×1
D3 dryer_tower	Towers	Dryer tower in service — Which of the two desiccant towers is in drying service; the other tower is regenerating. The controller alternates the towers on the tower period for as long as the unit delivers air. When the unit stops delivering, the changeover stops as well and the signal holds its last state.		—	10 s	alternating	1	1	9 uint16 ×1
D4 regulator_contact	MPG	Pressure regulator contact — Contact of the pneumatic pressure regulator on the panel. Its resting state means that the regulator sees enough pressure in the line. The contact drops out when the regulator calls for load, which is the pneumatic backup of the electronic regulation.		—	10 s	0	1	1	10 uint16 ×1
D5 low_pressure_switch	LPS	Low-pressure switch — Hardware pressure switch on the distribution line, wired straight to the controller. It closes when the line pressure falls below its fixed setting and opens again once the pressure has recovered past the reset point. Because it is independent of the pressure transducer, it still reports a collapsing line if the transducer fails.		—	10 s	0	0	0	11 uint16 ×1
D6 purge_switch	Pressure	Purge pressure switch — Pressure switch in the purge line of the desiccant dryer. Its active state reports that the off-line tower has discharged as expected. The controller watches it around every tower changeover to confirm that the dryer is regenerating.		—	10 s	1	1	1	12 uint16 ×1
D7 oil_level_ok	Oil_level	Oil level switch — Float switch in the oil separator vessel. Its resting state means that the oil level is above the minimum mark. It drops out when the level has fallen far enough to put the lubrication of the airend at risk.		—	10 s	1	1	1	13 uint16 ×1

Tag	MetroPT column	Description	Unit	Range	Sample rate	Normal loaded	Normal unloaded	Normal off	Register
D8 flow_pulse	Caudal_in	Flow pulse — Pulse output of the flow sensor at the reservoir inlet. Each pulse stands for a fixed volume of air entering the reservoirs, so the pulse rate follows the delivered flow. A signal that stops changing while the unit delivers points at the sensor or its wiring rather than at the air circuit.	—	—	10 s	1	1	1	14 uint16 ×1
T2 ambient_temperature	—	Ambient temperature — Temperature of the cooling air at the inlet of the enclosure. It fixes the cooling capacity available to the oil cooler and the aftercooler, so it enters the reading of every temperature on the unit. The controller also uses it to warn when the installation room leaves the operating limits.	°C	-20.0–60.0 °C (-4–140 °F)	10 s	2.0–40.0 °C (36–104 °F)	2.0–40.0 °C (36–104 °F)	2.0–40.0 °C (36–104 °F)	15 int16 ×100

9.3 Normal bands

A normal band is not a limit. Limits are the settings of chapter 4⁶⁶ and the messages of chapter 3⁶⁷; a band describes what a healthy unit reads while it is doing a particular thing. Bands are therefore given per machine state, because the same tag reads quite differently while the unit is loaded, while it is unloaded and while it stands off, and a value that is perfectly normal in one state is a fault in another.

The bands below were derived from a reference month of recorded operation of a healthy unit at reference conditions. Read them as the shape of healthy operation rather than as acceptance figures. A single reading outside its band is a reason to look, not a fault in itself. A reading that has crossed its band slowly over weeks says far more than one that happens to sit at the edge today, which is why 7.3.14⁶⁸ asks for the readings to be written down at every visit.

Normal bands and how they were derived

Tag	State	Normal band	Step	Source
P1	off	-0.1–0.0 bar (-1.45–0.00 psi)	0.1 bar (1.45 psi)	derived
P1	unloaded	-0.1–0.0 bar (-1.45–0.00 psi)	0.1 bar (1.45 psi)	derived
P1	loaded	8.8–10.4 bar (128–151 psi)	0.1 bar (1.45 psi)	derived
P2	off	8.1–9.4 bar (117–136 psi)	0.1 bar (1.45 psi)	derived
P2	unloaded	9.0–10.0 bar (131–145 psi)	0.1 bar (1.45 psi)	derived
P2	loaded	8.6–10.1 bar (125–146 psi)	0.1 bar (1.45 psi)	derived
P3	off	8.1–9.4 bar (117–136 psi)	0.1 bar (1.45 psi)	derived
P3	unloaded	9.0–10.0 bar (131–145 psi)	0.1 bar (1.45 psi)	derived
P3	loaded	-0.1–0.0 bar (-1.45–0.00 psi)	0.1 bar (1.45 psi)	derived
P4	off	-0.1–0.0 bar (-1.45–0.00 psi)	0.1 bar (1.45 psi)	derived
P4	unloaded	-0.1–0.0 bar (-1.45–0.00 psi)	0.1 bar (1.45 psi)	derived
P4	loaded	-0.1–0.0 bar (-1.45–0.00 psi)	0.1 bar (1.45 psi)	derived
P5	off	8.1–9.4 bar (117–136 psi)	0.1 bar (1.45 psi)	derived
P5	unloaded	9.0–10.0 bar (131–145 psi)	0.1 bar (1.45 psi)	derived
P5	loaded	8.6–10.1 bar (125–146 psi)	0.1 bar (1.45 psi)	derived
T1	off	50.0–61.0 °C (122–142 °F)	1 °C (34 °F)	derived
T1	unloaded	51.0–64.0 °C (124–147 °F)	1 °C (34 °F)	derived
T1	loaded	48.0–61.0 °C (118–142 °F)	1 °C (34 °F)	derived
I1	off	0.0–0.1 A	0.1 A	derived
I1	unloaded	3.6–4.0 A	0.1 A	derived
I1	loaded	5.8–6.3 A	0.1 A	derived
D1	off	1	1	derived
D1	unloaded	1	1	derived
D1	loaded	0	1	derived
D2	off	0	1	derived

⁶⁶ Chapter 4, Programmable settings

⁶⁷ Chapter 3, Controller CTRL-7

⁶⁸ Section 7.3.14, Recording the work and resetting the service message

Tag	State	Normal band	Step	Source
D2	unloaded	0	1	derived
D2	loaded	1	1	derived
D3	off	1	1	derived
D3	unloaded	1	1	derived
D3	loaded	alternating	1	derived
D4	off	1	1	derived
D4	unloaded	1	1	derived
D4	loaded	0	1	derived
D5	off	0	1	derived
D5	unloaded	0	1	derived
D5	loaded	0	1	derived
D6	off	1	1	derived
D6	unloaded	1	1	derived
D6	loaded	1	1	derived
D7	off	1	1	derived
D7	unloaded	1	1	derived
D7	loaded	1	1	derived
D8	off	1	1	derived
D8	unloaded	1	1	derived
D8	loaded	1	1	derived
T2	off	2.0–40.0 °C (36–104 °F)	1 °C (34 °F)	authored
T2	unloaded	2.0–40.0 °C (36–104 °F)	1 °C (34 °F)	authored
T2	loaded	2.0–40.0 °C (36–104 °F)	1 °C (34 °F)	authored

9.4 Spare parts and consumables

Order by the part code below and quote the serial number of the unit, which is stamped on the identification plate on the frame. The consumables are the parts named by the procedures in 7.3⁶⁹; keeping one set of them on site is what decides whether a service takes an afternoon or a fortnight.

Spare parts and consumables

Code	Part	Unit	Description
CAU7-AF-01	Air intake filter element		—
CAU7-OF-01	Oil filter element		—
CAU7-SE-01	Oil separator element		—
CAU7-OIL-46	Synthetic screw compressor oil	L	Synthetic screw compressor oil of the viscosity grade ISO VG 46, ordered per litre.
CAU7-DC-35	Desiccant charge for one tower	kg	—
CAU7-PS-01	Purge silencer		—
CAU7-DV-01	Condensate drain valve service kit		—
CAU7-PV-01	Dryer purge valve service kit		—
CAU7-CE-01	Coupling element		—
CAU7-CK-01	Cooler cleaning kit		—
CAU7-LS-01	Oil level switch		—

10 Appendix

10.1 Glossary

- **Airend** — the screw element in which the air is compressed.
- **Blow-down** — the venting of the separator vessel when the unit unloads or stops.
- **Changeover** — the dryer's switch between the drying and the regenerating tower.
- **Cut-in** — the line pressure at which the controller loads the unit.
- **Cut-out** — the line pressure at which the controller unloads the unit.
- **Cyclonic separator** — the vessel that spins condensate out of the cooled air.
- **Desiccant** — the porous material in the towers that dries the air.
- **Load** — the state in which the intake valve is open and air is delivered.
- **Minimum-pressure valve** — the valve that holds vessel pressure so the oil circulates.
- **Normal band** — the range a reading covers on a healthy unit in one state.
- **Pressure dew point** — the temperature at which the delivered air, at working pressure, condenses.
- **Purge** — the small flow of dried air that regenerates the resting tower.
- **Reference conditions** — the stated inlet and cooling conditions the ratings refer to.
- **Run-on** — the time the motor keeps turning unloaded after cut-out.
- **Scavenge line** — the line returning separated oil to the airend.
- **Separator element** — the element in the separator vessel that takes oil out of the air.
- **Service message** — a message reporting that a maintenance task has become due.
- **Shutdown** — a message that stops the unit and inhibits a start until it is reset.
- **Shutdown warning** — a message reporting that a shutdown limit is close; the unit runs on.
- **Unload** — the state in which the intake valve is closed and no air is delivered.
- **Warning** — a message reporting a reading outside its band; the unit runs on.

10.2 Revision history

Every issue of this manual is listed below.

Revision	Date	Change
1.0	2026-01-15	First issue

10.3 Licence and disclaimer

This manual is a work of fiction. The CAU-7 Compressed-Air Unit and the CTRL-7 controller do not exist: no such machine, component, part code or setting was ever built or sold, and nothing here is derived from any real product's documentation. It is reference material for a fault-diagnosis research project and must never be used to install, operate, service or repair real equipment.

This manual is licensed under the Creative Commons Attribution 4.0 International Licence (CC BY 4.0), <https://creativecommons.org/licenses/by/4.0/>. Attribute it to *Meddle S.r.l.* You may share and adapt it, including commercially, provided you give credit, link to the licence and state whether changes were made.

10.4 Data credit

The normal bands of 9.3⁷⁰ and the reference operation behind them were derived from a public recording of a real air production unit: Davari, N., Veloso, B., Ribeiro, R., & Gama, J. (2021). *MetroPT-3 Dataset*. UCI Machine Learning Repository. <https://doi.org/10.24432/C5VW3R>. That dataset is licensed under CC BY 4.0.

Only statistical summaries were taken from it: the bands and the cycle figures in chapter 9⁷¹. The machine described here is not the machine that was recorded, and no text or drawing came from the dataset or its documentation.

Air intake filter

Dry panel filter that cleans the air the airend draws in.

Intake and unloading valve

Combined intake and unloading valve that opens the air path when the controller loads the unit.

Screw compression element

Oil-injected single-stage screw element that compresses the intake air.

Drive motor

⁷⁰ Section 9.3, Normal bands

⁷¹ Chapter 9, Technical data and signal list

Fixed-speed three-phase motor that drives the compression element.

Motor coupling

Flexible coupling between the motor shaft and the compression element.

Oil separator vessel

Pressure vessel that holds the oil charge and carries the separator element that returns oil to the circuit.

Minimum-pressure valve

Non-return valve that keeps a minimum pressure in the separator vessel so that oil keeps circulating.

Blow-down valve

Solenoid valve that vents the separator vessel to the intake silencer whenever the unit unloads or stops.

Oil cooler

Air-to-oil heat exchanger that removes the heat of compression from the oil charge.

Aftercooler

Air-to-air heat exchanger that cools the delivered air before it reaches the cyclonic separator.

Cooling fan

Fan that draws cooling air through the enclosure and across both coolers.

Thermostatic valve

Valve that bypasses the oil cooler until the oil has reached its working temperature.

Oil filter

Spin-on filter that removes wear particles from the oil before it re-enters the compression element.

Cyclonic separator

Centrifugal separator that removes the condensate that forms in the aftercooler.

Automatic condensate drain

Automatic drain valve that empties the cyclonic separator into the condensate connection.

Desiccant tower 1

First of the two desiccant towers; it dries the air while the second one regenerates.

Desiccant tower 2

Second of the two desiccant towers; it dries the air while the first one regenerates.

Dryer changeover valves

Valve block that swaps the drying and the regenerating tower at the end of each tower period.

Dryer purge valve

Valve that releases the regeneration air of the off-line tower to the purge silencer.

Purge silencer

Silencer that damps the noise of the purge air leaving the dryer.

Air reservoirs

Two pressure vessels that store dried air and damp the demand of the installation.

Reservoir isolation valve

Manual valve that separates the reservoirs from the distribution line for service work.

Pneumatic panel

Panel that carries the line pressure transducer, the pressure regulator and the low-pressure switch.

Safety valve

Spring-loaded relief valve on the separator vessel that protects the unit against overpressure.

CTRL-7 controller

Electronic controller that regulates the unit, evaluates the messages and drives the display.

Oil level switch

Float switch in the separator vessel that reports whether the oil level is above the minimum mark.

Reservoir inlet flow sensor

Pulse sensor at the reservoir inlet that reports the volume of air entering the reservoirs.

Manual, fault catalog and synthetic data: CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>). Code: Apache-2.0.

MetroPT-3: Davari, N., Veloso, B., Ribeiro, R., & Gama, J. (2021). *MetroPT-3 Dataset*. UCI Machine Learning Repository. <https://doi.org/10.24432/C5VW3R>. Licensed CC BY 4.0, redistributed unchanged.

CAU-7 Compressed-Air Unit and CTRL-7 are fictional. Do not use this manual to operate or maintain real equipment.